



Robotics in Manufacturing

(Spring, 2026)

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Brief Information

Course Title: SMMG6000M – Robotics in Manufacturing

Instructor: Pengcheng HU, pengchenghu@hkust-gz.edu.cn, W2-610

Grading policy: final project (70%), attendance (10%), in-class performance and quiz (20%).

Course materials:

- Robotics, Mechanics and Control (Fourth Edition), by John Craig.
- Robot Modeling and Control (Second Edition), by Mark W. Spong, et al.
- Robotics in Manufacturing, by Carl Keynes

Teaching Assistant: Xingke Xia, xxia260@connect.hkust-gz.edu.cn

Yifan Liao, yliao206@connect.hkust-gz.edu.cn



Course Arrangements

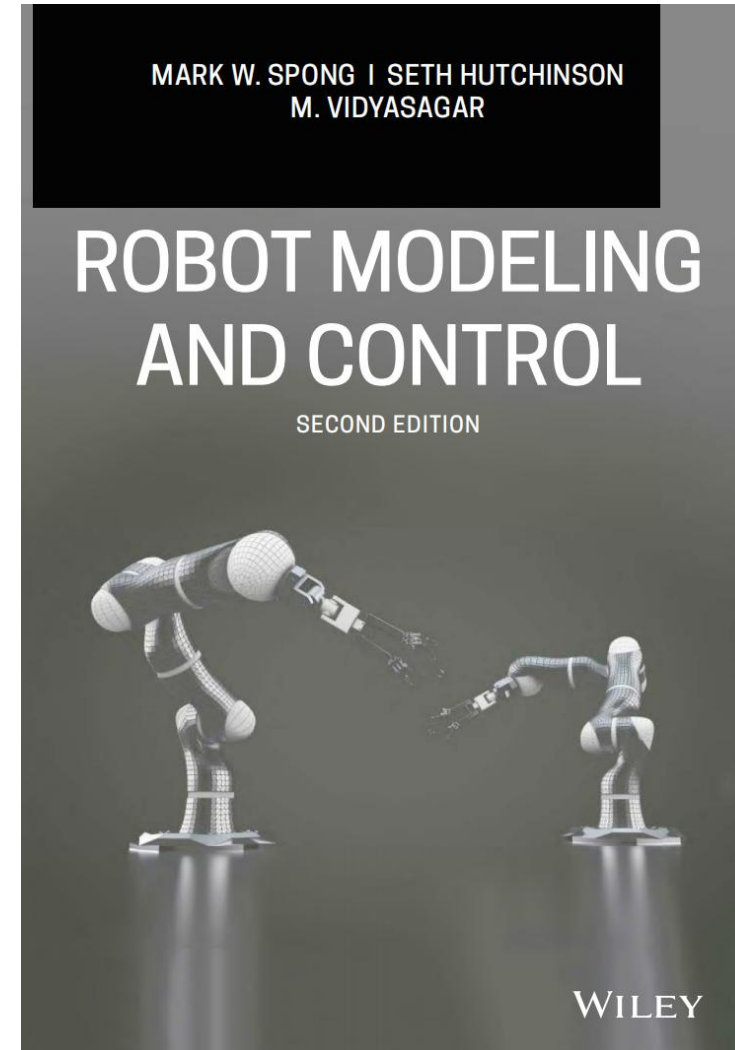
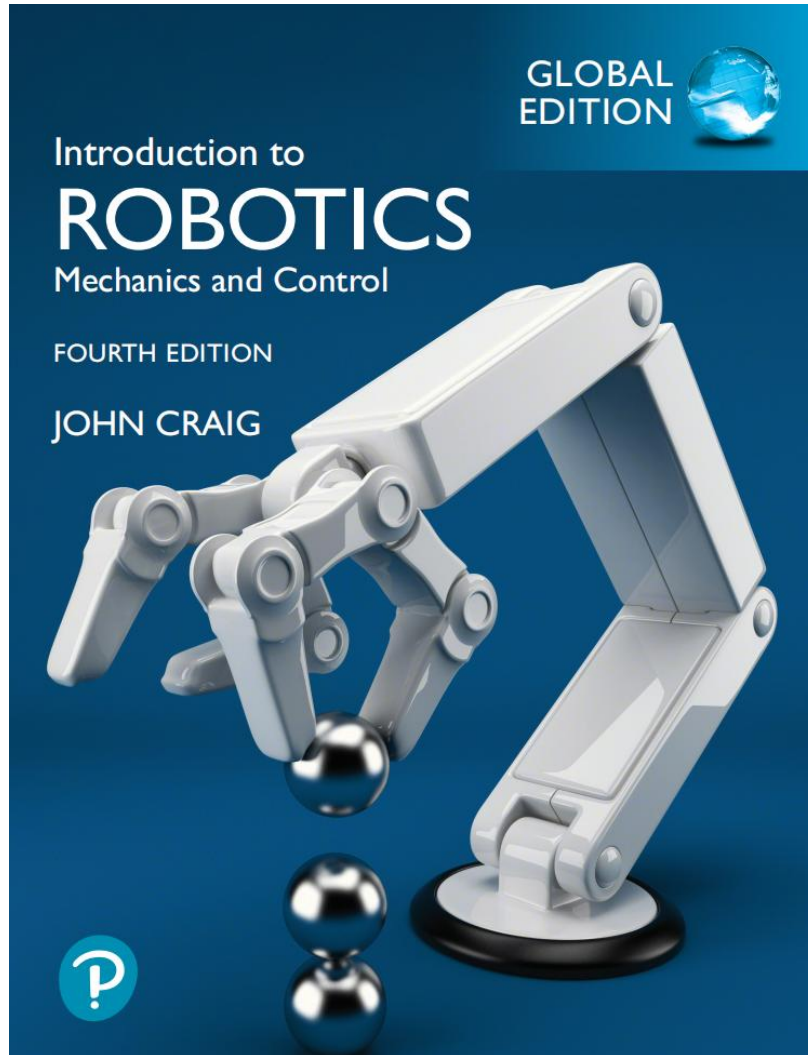
Week	S	M	T	W	T	F	S	Events	General Holidays
January, 2026									
					1	2	3		1-3 New Year's Day
	4	5	6	7	8	9	10	9 Validation Period Starts - All UG/PG Students	
	11	12	13	14	15	16	17	15 Class Enrollment Starts - All UG/PG Students	
1	18	19	20	21	22	23	24	22 Last day of Winter Term	
1	25	26	27	28	29	30	31	23 Spring Term commences	
February									
2	1	2	3	4	5	6	7	23/1-5/2 Add/Drop Period	
3	8	9	10	11	12	13	14	11-23 Winter Break for UG students (no centrally scheduled PG classes)	15-23 Spring Festival
	15	16	17	18	19	20	21		
3	22	23	24	25	26	27	28	28 Make-up day for Monday classes	
March									
4	1	2	3	4	5	6	7	6/2-4/3 Extended Drop Period - For PG Courses Only	
5	8	9	10	11	12	13	14		
6	15	16	17	18	19	20	21		
7	22	23	24	25	26	27	28		
8	29	30	31						
April									
8				1	2	3	4		
9	5	6	7	8	9	10	11		4-6 Qingming Festival
10	12	13	14	15	16	17	18		
11	19	20	21	22	23	24	25		
12	26	27	28	29	30				
May									
						1	2		1-5 Labor Day
13	3	4	5	6	7	8	9	11 Last day of Spring Term classes	
13	10	11	12	13	14	15	16	12-16 Study Break	
	17	18	19	20	21	22	23	18-29 Spring Term Examinations	
	24	25	26	27	28	29	30	29 Last day of Spring Term	
	31								

In total 13 weeks, 42 teaching hrs

- 11 Weeks lectures
- 1 Week lab (lab tour and programming of industrial robot)
- 1 Week project and presentation

Academic calendar in 2026 Spring Term

References





Basic theories of industrial robotics as well as advanced manufacturing methods utilizing industrial robots.

- Part I will introduce the fundamental theories of industrial robots.
- Part II will focus on **applications of industrial robot in specific manufacturing scenarios**, such as machining, additive manufacturing, and multi-robot collaboration, exploring advanced manufacturing methods centered around robots.

10 Chapters in total:

- Chapter 1~6: Fundamental theories of industrial robotics.
- Chapter 7~10: Integration of industrial robot with different types of manufacturing.



Syllabus

Chapter 1: Introduction to Robotics

- Introduction of robotics
- Industrial robot in manufacturing

Chapter 2: Spatial Descriptions and Transformations

- Descriptions: positions, orientations and frames
- Mappings: changing descriptions from frame to frame
- Operators: translations, rotations, and transformations
- Transformation arithmetic and equations
- Transformation of free vectors

Chapter 3: Manipulator Kinematics

- Link and link-connection description
- Manipulator kinematics

Chapter 4: Inverse Manipulator Kinematics

- Existence of Solutions
- Pieper's Solution for Triaxial Intersection
- Standard Coordinate System

Chapter 5: Jacobians: Velocity and Static Forces

- Velocity of manipulator
- Jacobians
- Static forces in manipulators
- Cartesian transformation of velocities and static forces

Chapter 6: Path and Trajectory Planning

- Configuration space
- Path planning for $Q=\mathbb{R}^2$
- Artificial potential field
- Sampling-based methods
- Trajectory planning



Chapter 7: Robotic Vision

- 3D vision for navigation and grasping
- Image processing and analysis of robotics

Chapter 8 - Robotic milling

- Process of milling
- Robotic milling technology

Chapter 9 - Robotic DED manufacturing

- Processing of DED manufacturing
- Robotic DED technology

Chapter 10 – Robotic carbon fiber placement

- Process of carbon fiber placement
- Robotic carbon fiber placement technology



Chapter 1 Fundamentals

1. Introduction of robotics

2. Industrial robot in manufacturing



What's the Robot?



Unitree Humanoid Robot G1

<https://www.youtube.com/watch?v=GzX1qOIO1bE>



Boston Dynamics Atlas

<https://www.youtube.com/watch?v=YIhzUnvi7Fw>



Figure AI Humanoid Robot

https://www.youtube.com/watch?v=zMg5_Dwz300



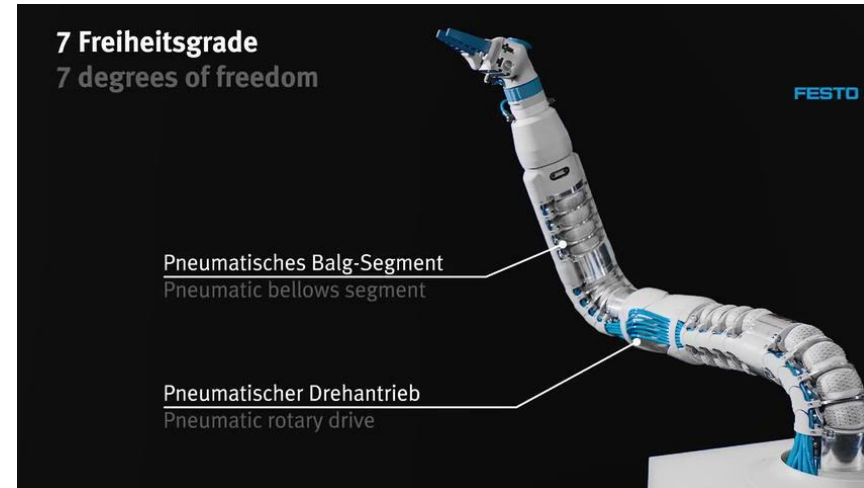
Boston Dynamic-spot mini quadruped robot

<https://www.youtube.com/watch?v=MG4PPkCyJig>

What's the Robot?



HONDA-ASIMO humanoid robot



FESTO - Elephant Trunk Bionic Robot



FESTO-Bat Bionic Robot

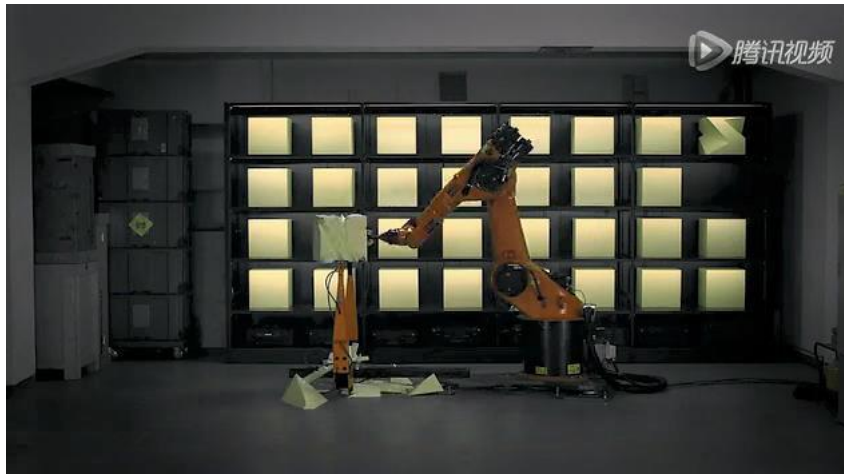


FESTO - Butterfly Bionic Robot

What's the Robot?



ABB-Industrial Robots



KUKA-Industrial Robots



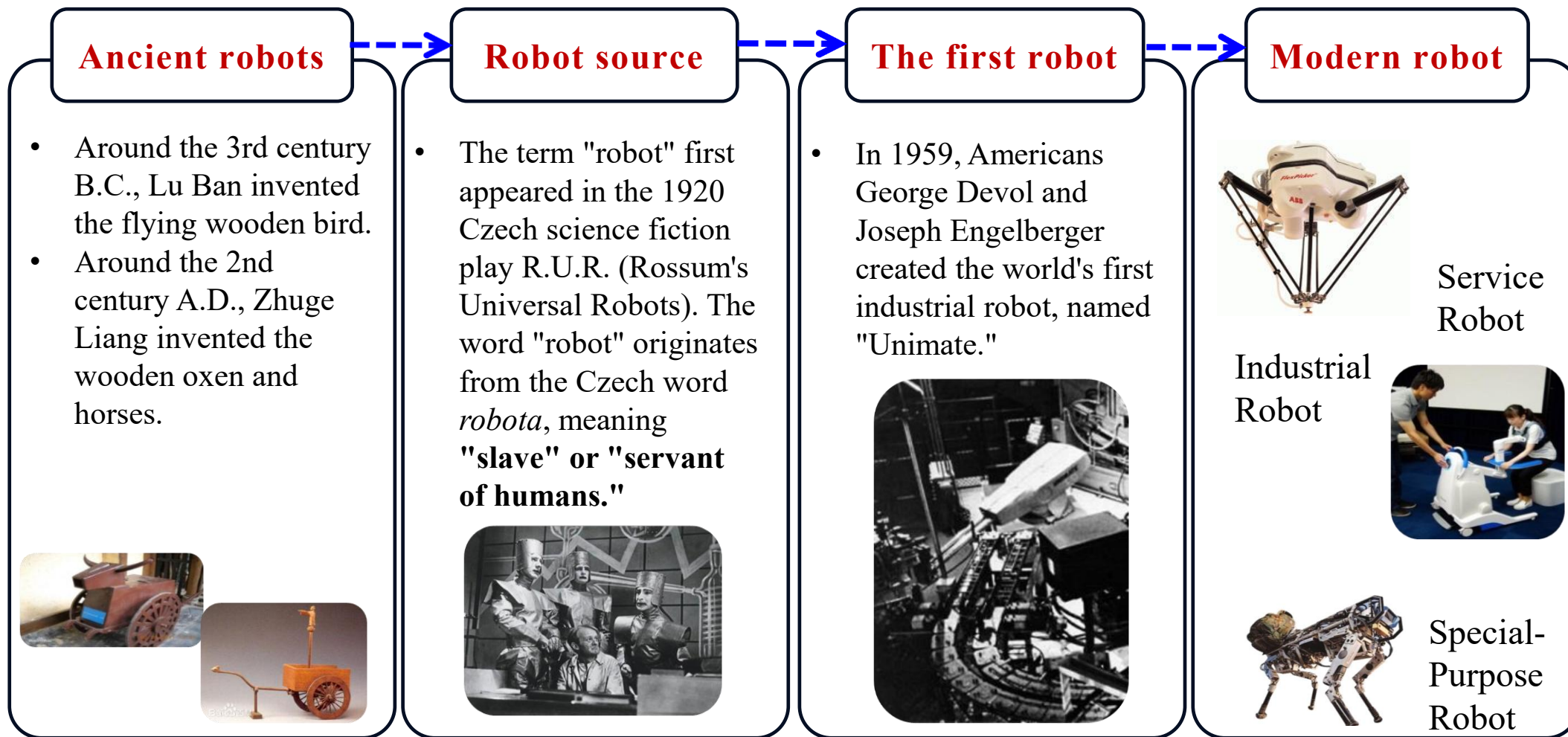
SiYi - Rehabilitation Robot



Russian - Skybot Robot

What's the Robot?

Development of Robotics



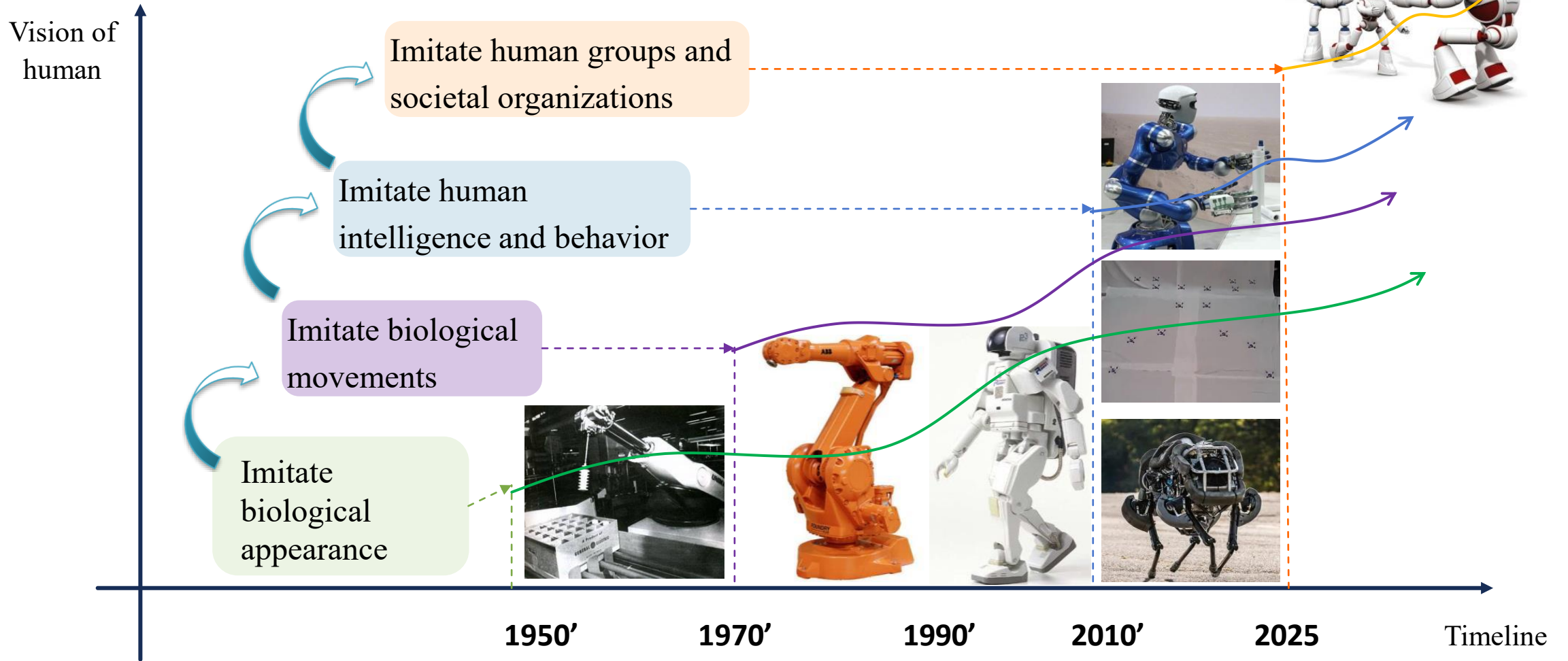
Purpose of creating and inventing robots is to **replace, expand, and serve humans.**

Human dream is to make robots a **friendly partner** for humans.

What's the Robot?



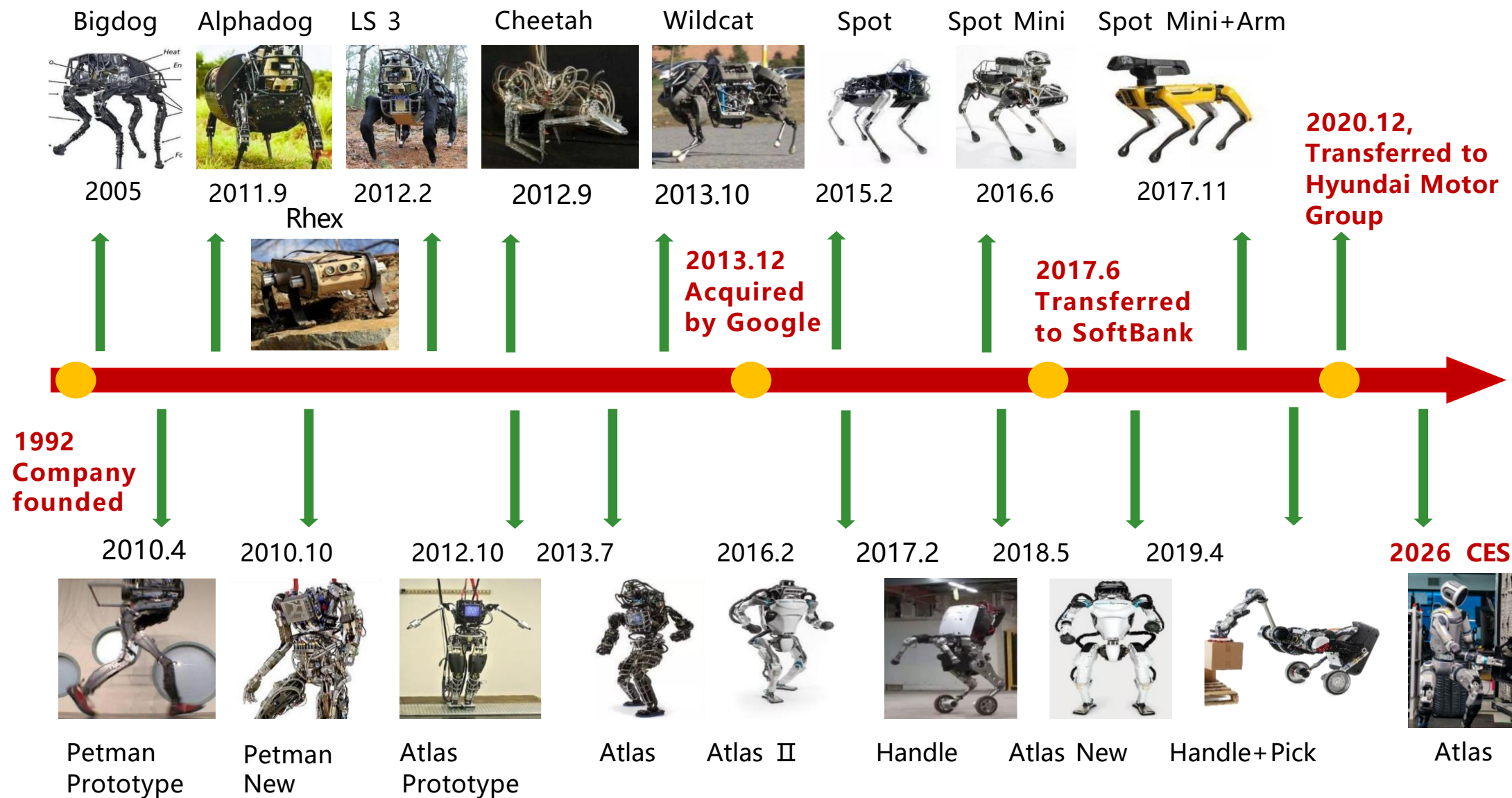
Roadmap of Modern Robot Development



Human expectations for robots have increasingly focused on their structure, functions, behaviors, and intelligence

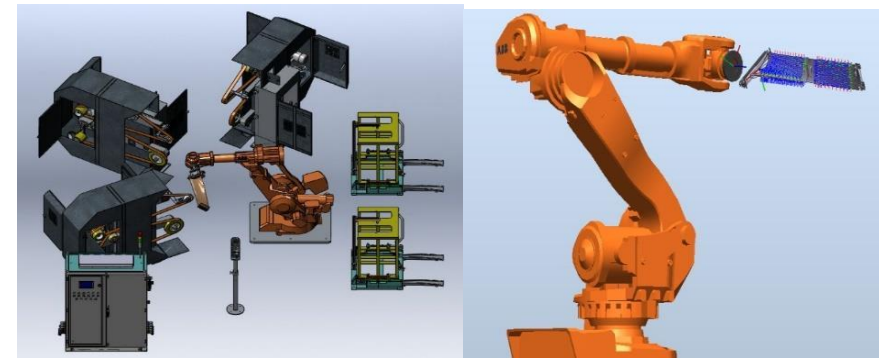
What's the Robot?

History of Boston Dynamic



Why Robots are Needed

- ❑ In environments that are dirty, dangerous, or tedious, where humans find it difficult or impossible to work, robots are needed to complete these tasks.



Dirty, polluted, and hazardous working conditions



Dangerous environments



Repetitive and tedious jobs



Definition of Robotics

❑ **International Organization for Standardization (ISO) Definition:** A robot is an automatic, position-controlled, multi-functional manipulator with programmable capabilities. It has several axes and can process various materials, parts, tools, and specialized devices through programmable operations to perform different tasks.

1	2	3	4
The action mechanism of a robot has functions similar to certain organs of humans or other biological entities (e.g., limbs, sensory organs).	Robots are flexible and adaptable , capable of performing a wide variety of tasks with dynamic operational programs.	Robots have varying degrees of intelligence , such as memory, perception, reasoning, decision-making, and learning.	Robots operate as independent and complete systems, capable of functioning without human intervention.



Definition of Robotics

- **Definition by the U.S. National Bureau of Standards (NBS):**

A robot is a mechanical device capable of being programmed to perform specific operational and movement tasks under automatic control.

- **Definition by the Robotics Industries Association (RIA):**

A robot is a programmable, multifunctional manipulator designed to move various materials, parts, tools, or specialized devices and execute a wide range of tasks through programmable actions.

- **Definition by the Japan Industrial Robot Association (JIRA):**

An industrial robot is a general-purpose machine equipped with memory devices and end-effectors, capable of rotating and completing various movements automatically to replace human labor.



Classification of Robots

Robot Classification by Application: Industrial Robots, Service Robots, Medical Rehabilitation Robots, Special-Purpose Robots, Exploration Robots, etc.



Industrial Robots



Service Robots



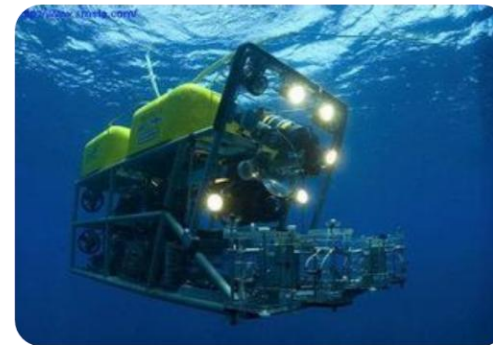
Medical Rehabilitation Robots



Defense and Military Robots



Space Exploration Robots



Resource Exploration Robots

Classification of Robots



Industrial robots



Service, mobile and medical robots

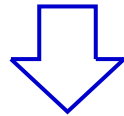


This course mainly focuses on industrial robots

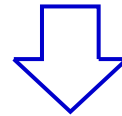
History of Industrial Robots



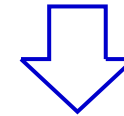
1960s



1980s



21Century



Teaching and Playback



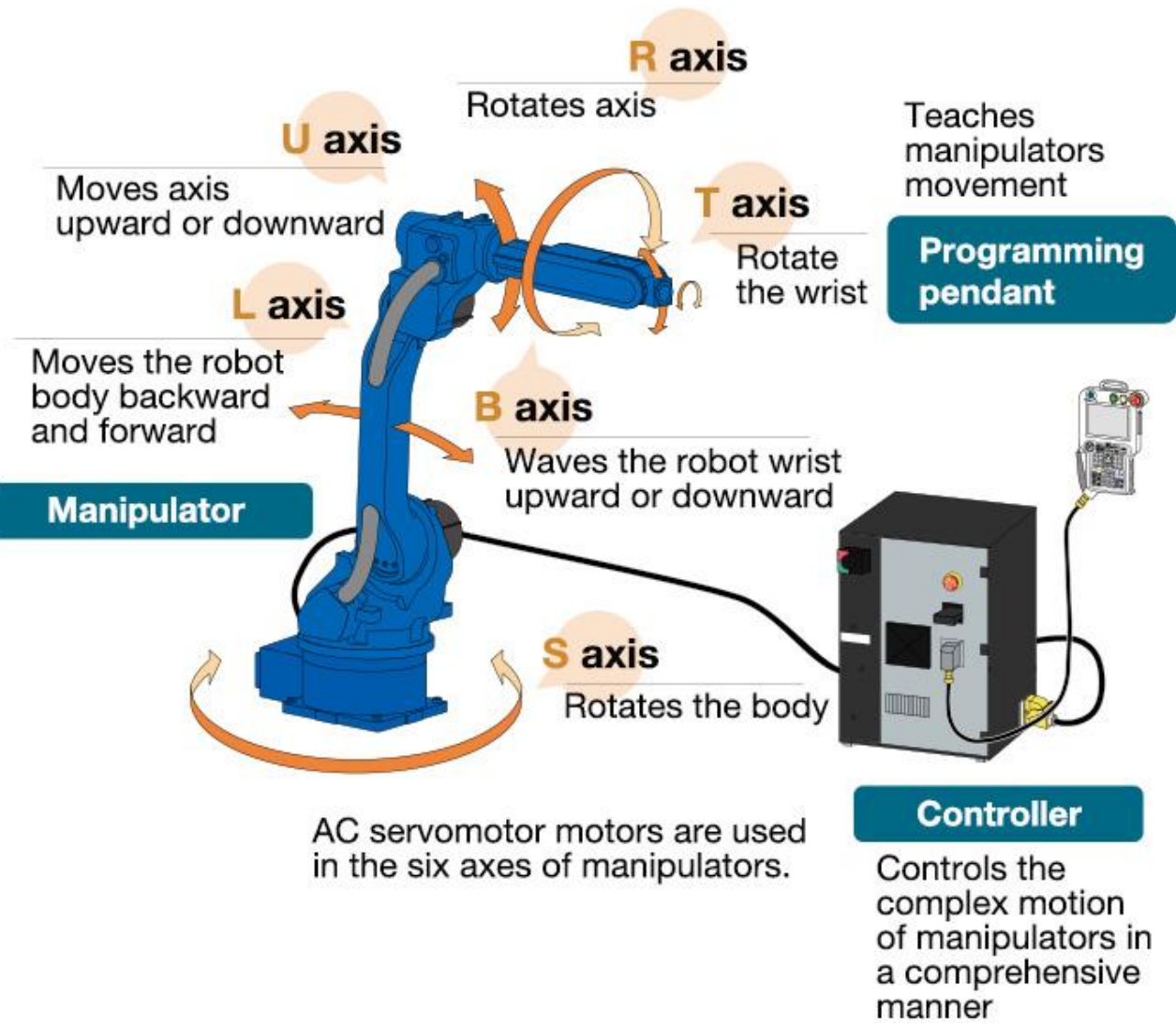
Perception-based Robots



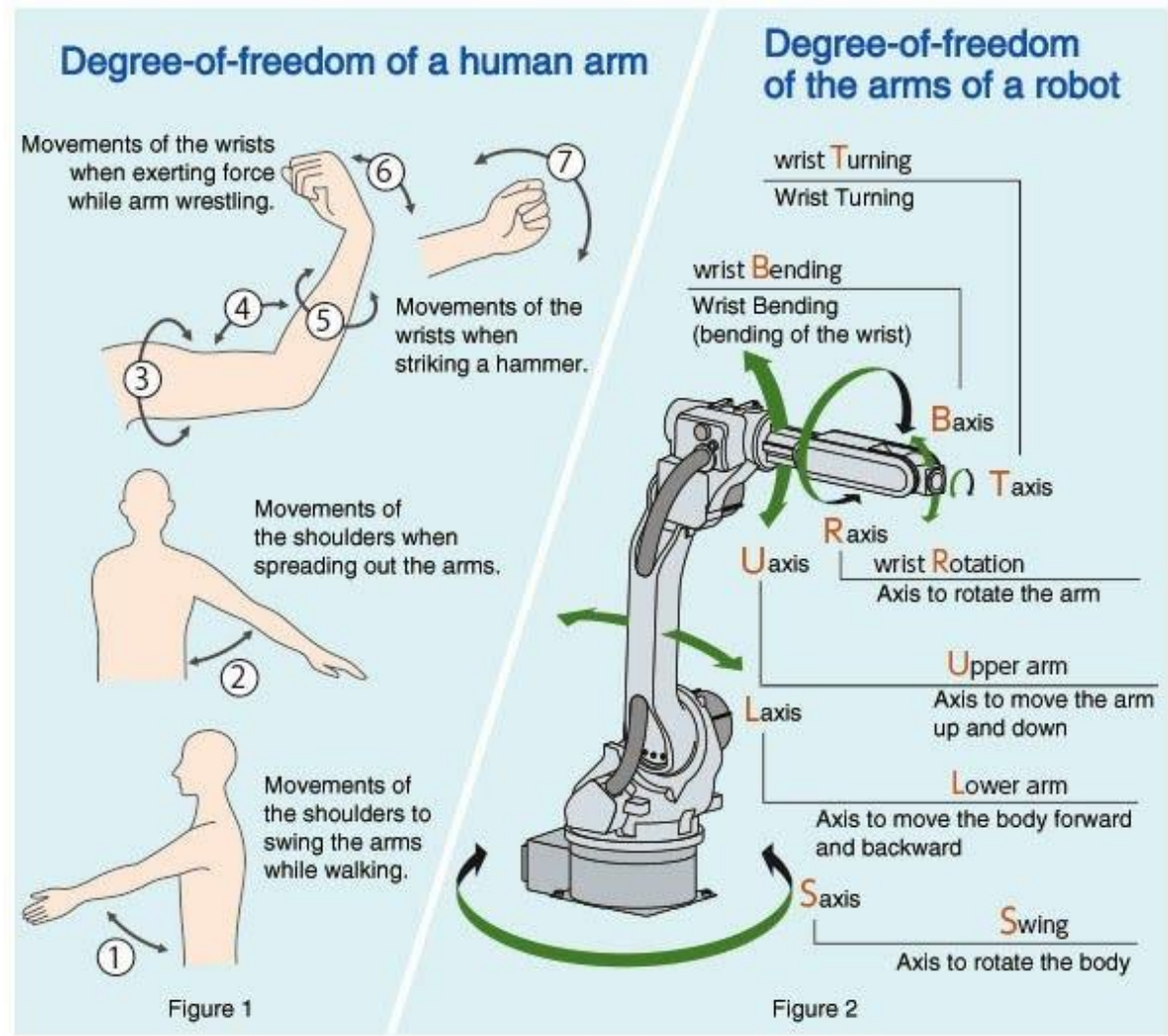
Intelligent Robots



Components of the Operating Arm System



Components of Industrial Robot

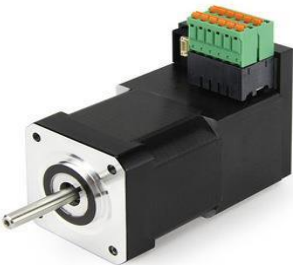


Industrial Robot vs. Human Arm

Components of the Operating Arm System

▣ Servomotor & Precision Reducer

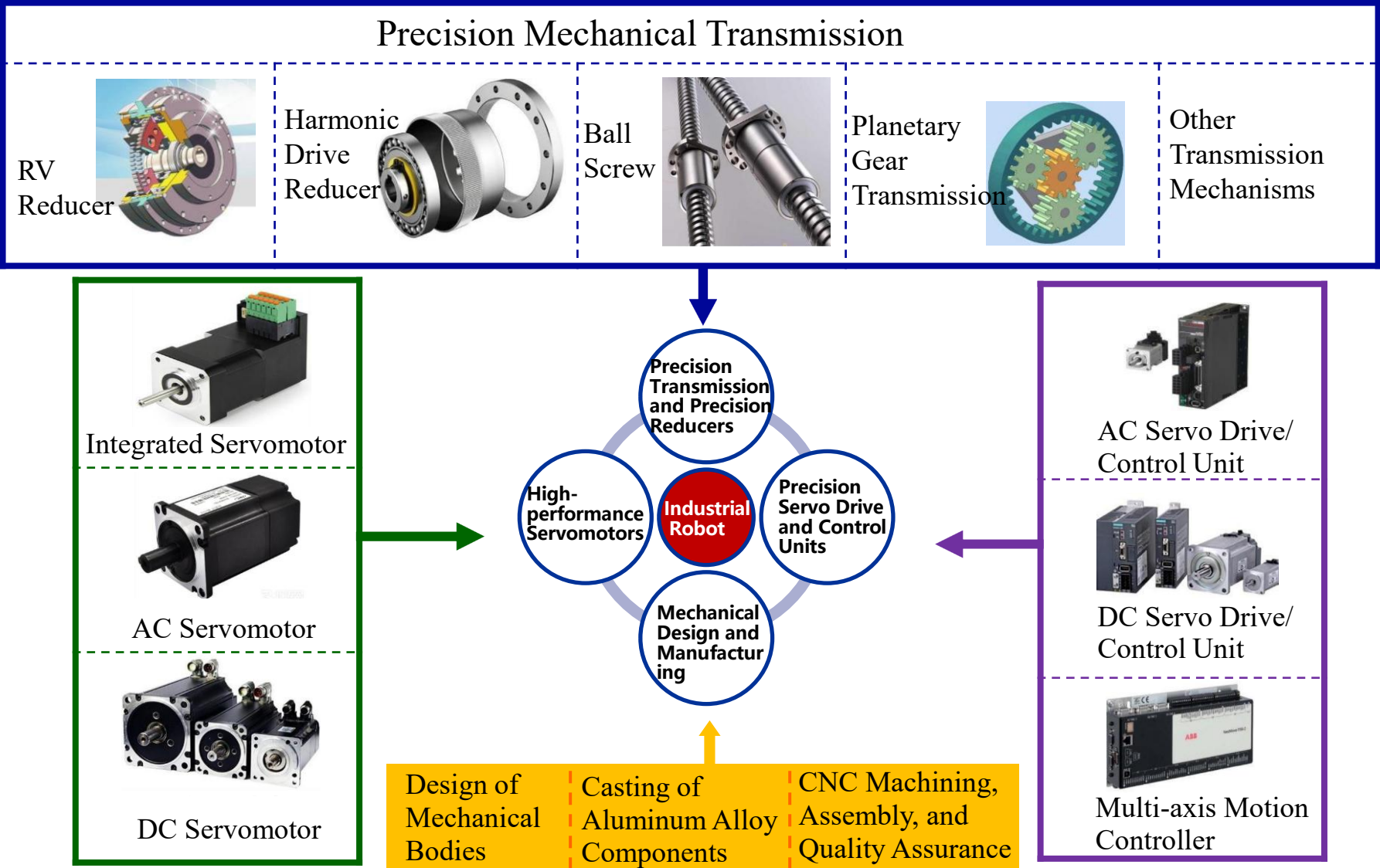
Precision Mechanical Transmission (Precision Reducers)			
RV Reducer		Harmonic Drive Reducer	
		Ball Screw	
			Planetary Gear Transmission and Other Components

Servomotors		
		
Integrated Servomotor	AC Servomotor	DC Servomotor

Precision reducers and AC/DC servomotors used in industrial robot manipulators.

Components of the Operating Arm System

Core components of the robotic arm



Components of the Operating Arm System

□ Representative Industrial Robot Products



Yaskawa Robot



KUKA Robot



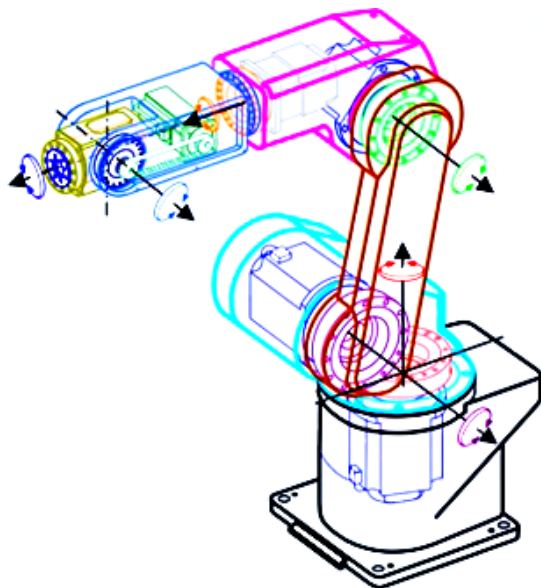
ABB Robot



HNC Robot

Components of the Operating Arm System

- Example of structural components



Axes J1 to J3 also use RV cycloidal gear reducers.

Axes J4 to J6 adopt harmonic drive gear mechanisms.

3D layout design of the MOTOMAN industrial robot manipulator's servo motor drive and mechanical transmission system.

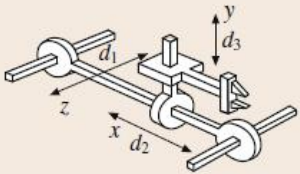
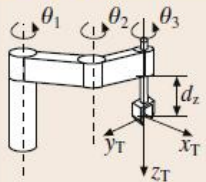
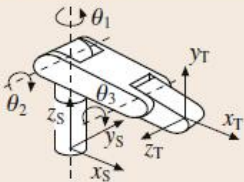
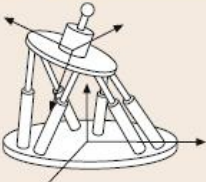
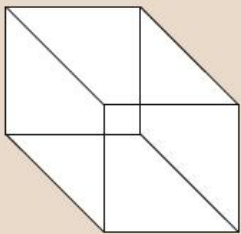
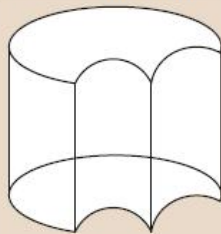
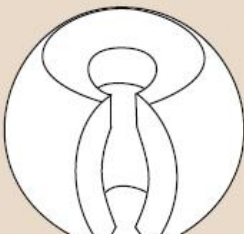
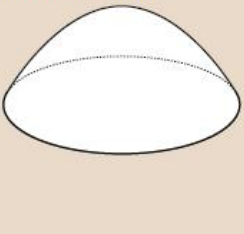
Basic Parameters and Operational Requirements of the Operating Arm

General Basic Parameters for Industrial Robotic Manipulators

- Workspace Geometry and Size
- Effective Payload
- Speed
- Degrees of Freedom
- Motion Accuracy

Workspace

Within the motion range of each joint, the workspace refers to the **geometric space formed by all the reachable positions** of the robotic manipulator's end-effector center point, given the robot's structure and configuration parameters.

Category	Gantry (or Cartesian)	SCARA	Articulate	Parallel
Robot main axes structure	 3 prismatic joints	 1 prismatic and 2(3) revolute joints	 3 revolute joints	 Typically with 3, 4 or 6 prismatic axis
Workspace shape				
Technical example				

Working spaces of typical industrial robot



Basic Parameters and Operational Requirements of the Operating Arm

Effective payload: the load-carrying or operating capacity of the robot's end-effector under the drive of its main components, excluding the self-load caused by the robot's own mass, inertia, friction, etc., during motion Payload is typically measured in kilograms as the maximum end load and is also related to the position of the load.

Motion accuracy: Refers to the deviation between the actual position reached by the center point of the robot's end-effector and the desired target position. This deviation is composed of the robot's mechanical precision and control precision. Evidently, the smaller the deviation, the higher the accuracy, and the more precise the achieved position.

Speed: the linear speed of the robot's end-effector center point movement and the angular speed of its posture changes. These speed parameters are determined by the robot's structure, the speed of each joint, and the parameters of the linkages.



Industrial Robot Example

ABB IRB 6660-130/3.1

Specifications

Brand	ABB
Model	IRB 6660-130/3.1
Type	Robot Arm
Axes	6
Payload	130 kg
Reach	3100 mm
Repeatability	0.11 mm
Weight	1910 kg

The ABB IRB 6660-130/3.1 robot is a 6-axis robot arm, it has a payload of 130 kg and a reach of 3100 mm. Manufacturing applications include Remote TCP.

Common applications of the ABB IRB 6660-130/3.1 include: Additive Manufacturing, Assembly, Dispensing, Finishing, Material Handling, Palletizing, Remote TCP.

The ABB IRB 6660-130/3.1 is manufactured by [ABB](#).

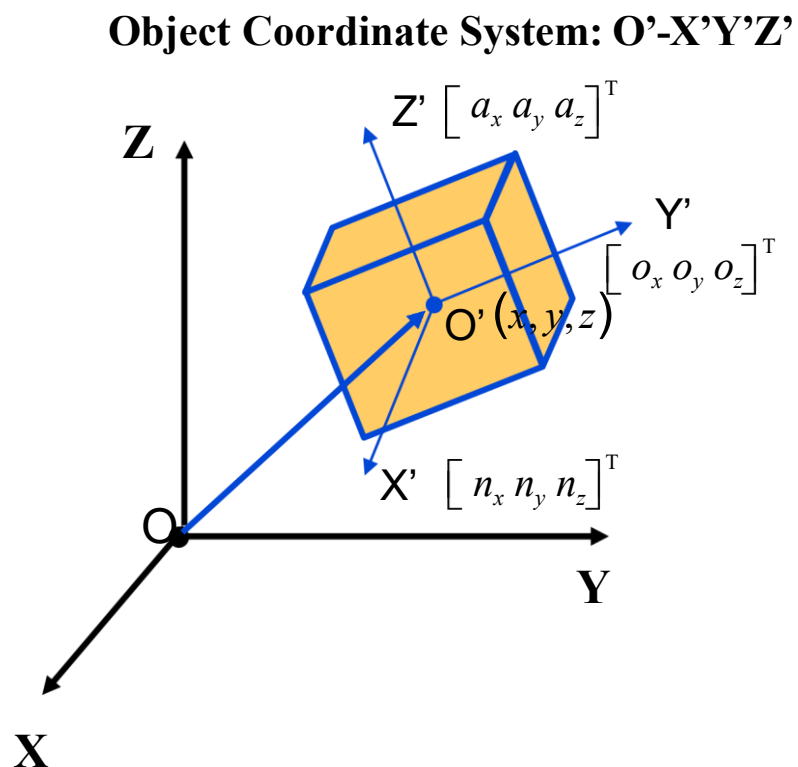
Use this [robot comparator](#) to find similar robots to the ABB IRB 6660-130/3.1.

Browse the full [RoboDK library](#).



Basic Parameters and Operational Requirements of the Operating Arm

□ Position and posture of the object



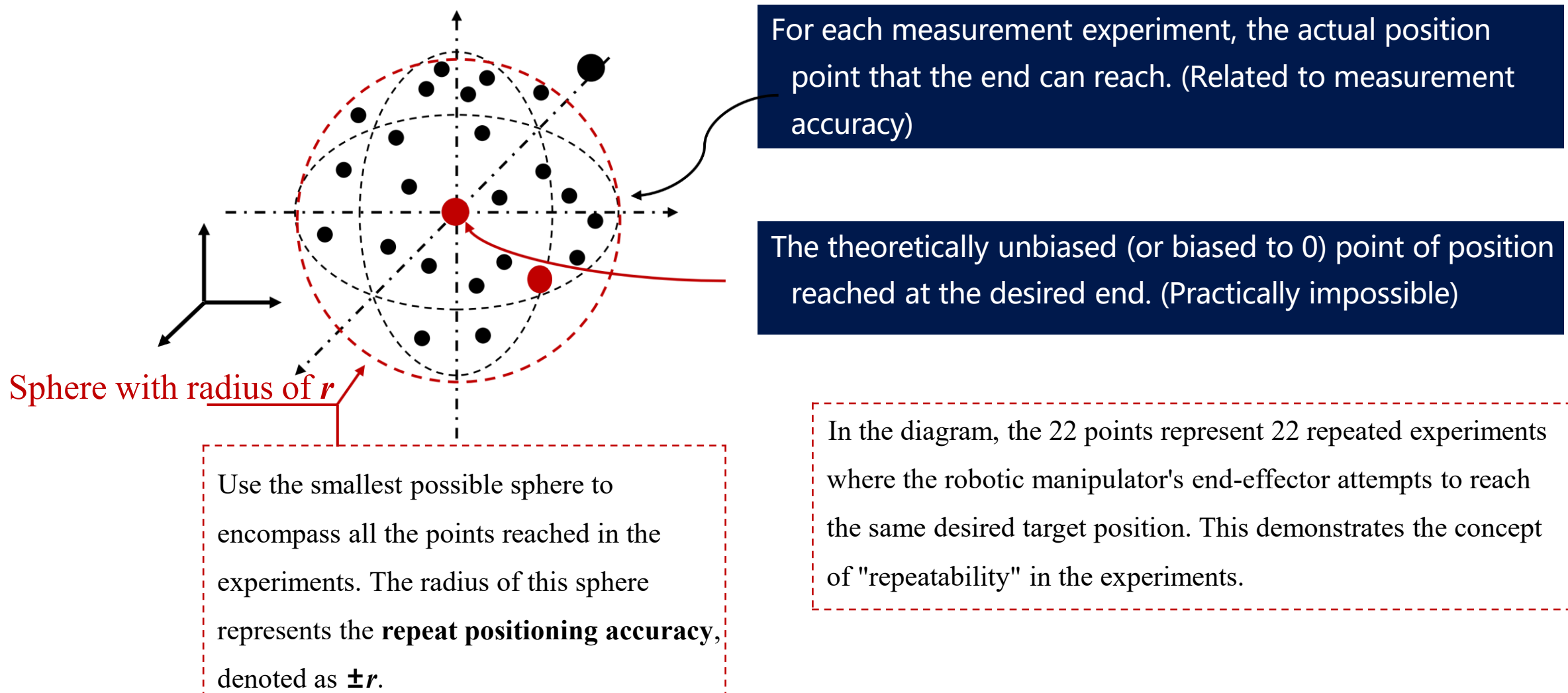
Object Position: (x, y, z)

Object Posture: The O'-X'Y'Z' coordinate system is represented by the posture matrix R in the base coordinate system O-XYZ:

$${}^O R_{O'} = \begin{bmatrix} n_x & o_x & a_x \\ n_y & o_y & a_y \\ n_z & o_z & a_z \end{bmatrix}$$

Basic Parameters and Operational Requirements of the Operating Arm

- Repeat positioning accuracy at the end of the robotic arm





Chapter 1 Introduction to Robotics

1. Introduction of robotics
- 2. Industrial robot in manufacturing**

Industrial Robot in Manufacturing

□ Typical industrial robotic arms

Welding



Spot Welding Robot



Arc Welding Robot

Painting



Painting Robot

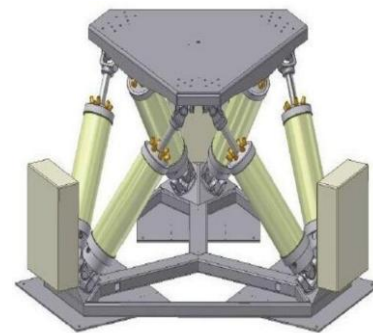
Assembly



SCARA Robot



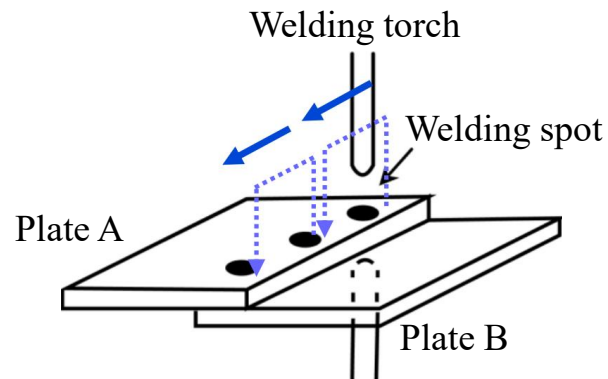
Stewart Platform (Parallel Mechanism Platforms)



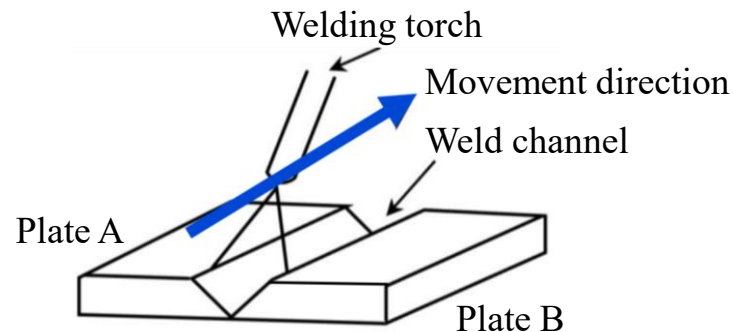
Industrial Robot in Manufacturing

□ Operating requirements for robotic arms

Spot Welding and Arc Welding

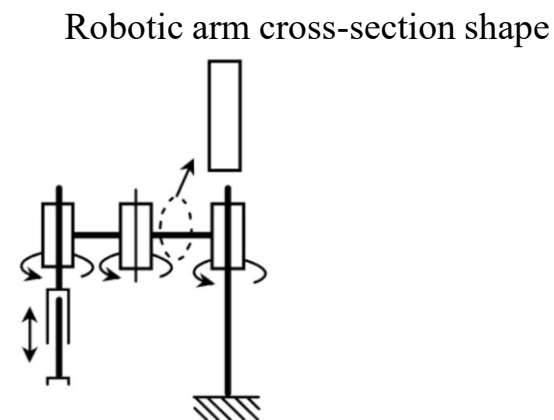
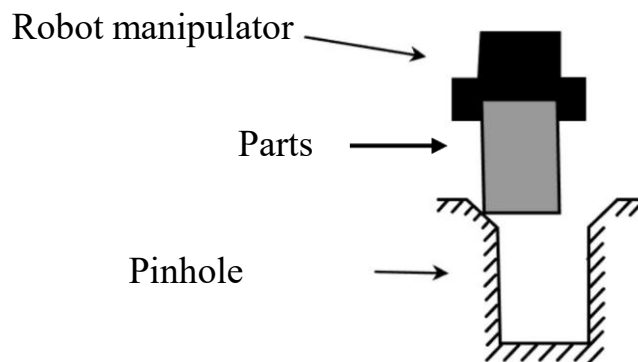


Robot arm carries torch on intermittent point-to-point trajectory



Robot takes torch on continuous trajectory

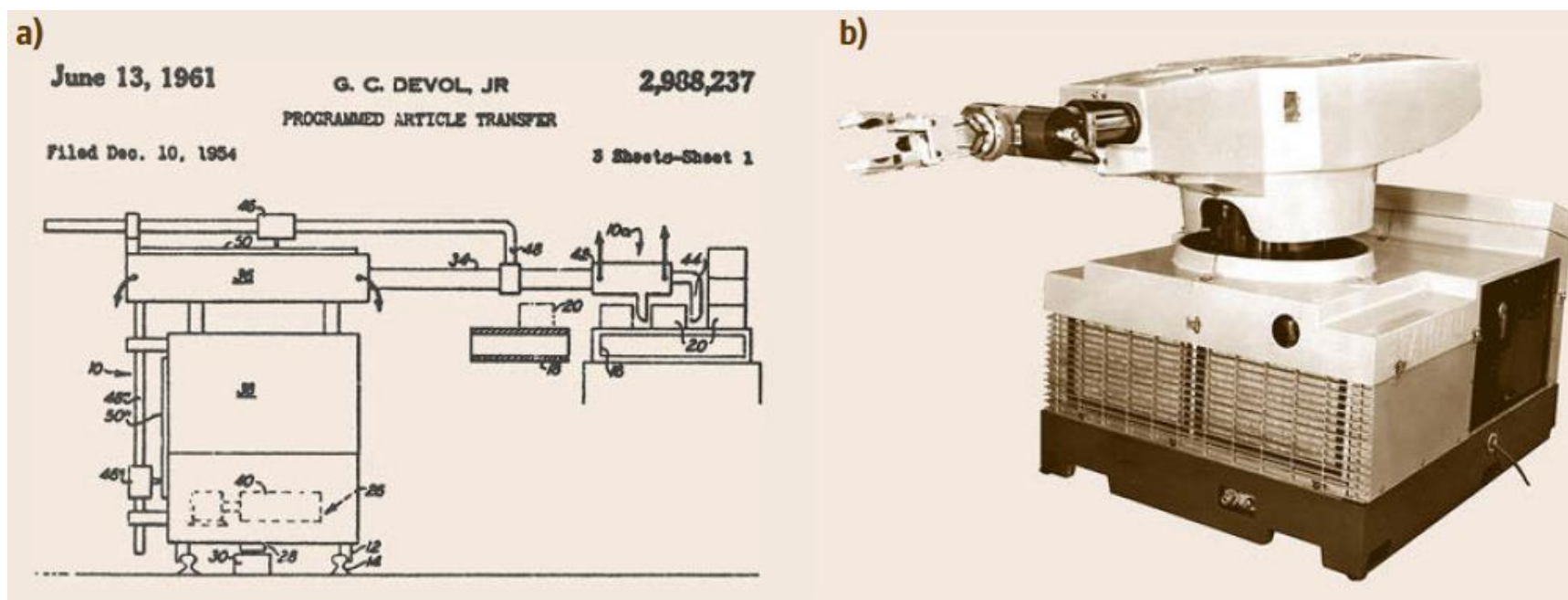
Pin-In-Hole Assembly



The robotic arm, equipped with a pin-gripping end-effector, must move precisely along the trajectory of the pinhole center to insert the pin into the hole

Industrial Robot in Manufacturing

❑ Short History of Industrial Robots in Manufacturing



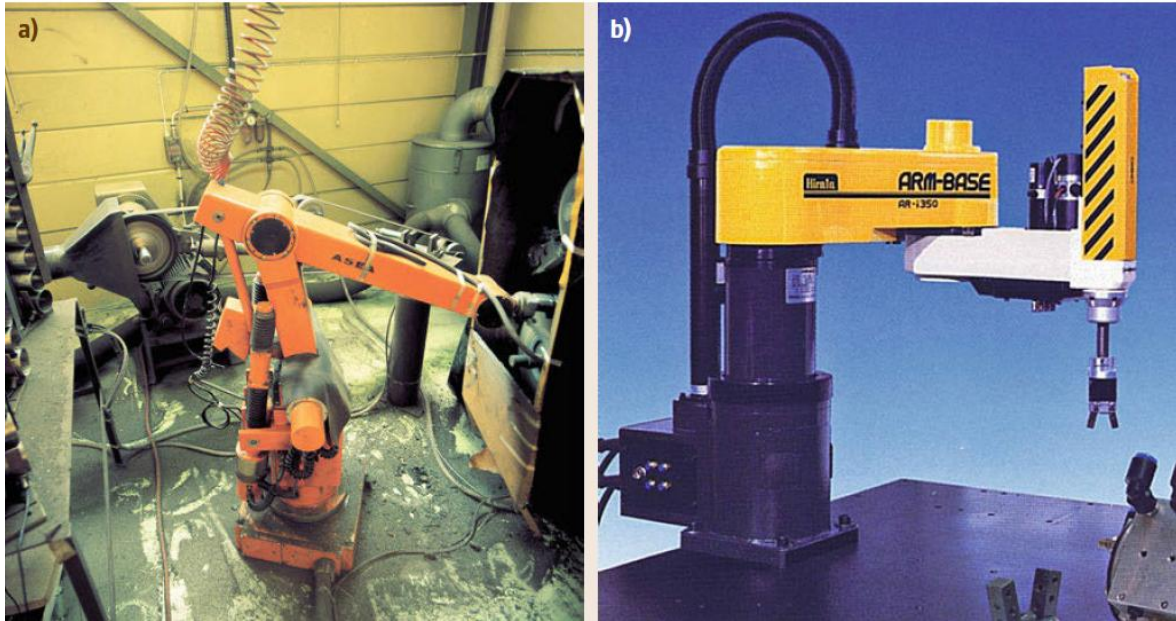
The invention of the industrial robot - Unimation, a fusion of the terms *universal* and *automation*

The first Unimation performed a rather simple handling task in 1961 at a General Motors plant; other car manufacturers followed (courtesy of Smithsonian Institution Archives, Washington DC).

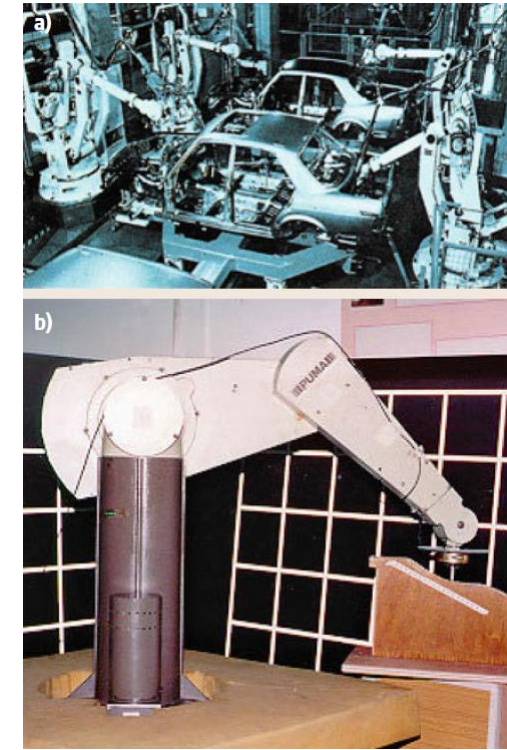


Industrial Robot in Manufacturing

❑ Short History of Industrial Robots in Manufacturing



- The all-electric (a) IRB-6 and (b) a SCARA-type kinematic**
- (a) First introduced in 1973, the IRB-6 was the first serially produced robot product, which combined all-electric-drives technology and a microcomputer for motion control and programming.
- (b) The selective compliance assembly robot arm (SCARA) is particularly suited for assembly tasks as it combines rigidity in the vertical axis and compliance in the horizontal axis.

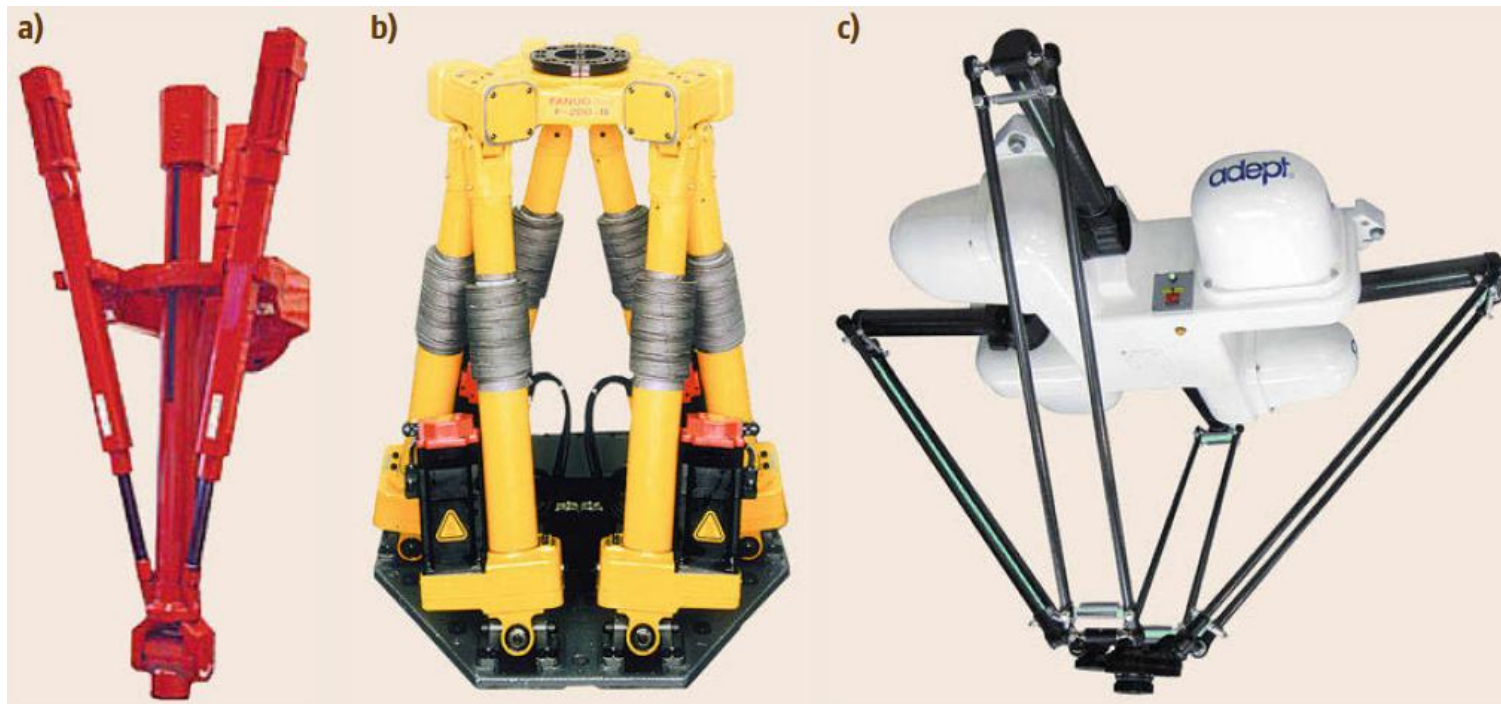


The KUKA IR 601/60 (a) and the Unimation PUMA (programmable universal machine for assembly) 560 (b)

- (a) In 1978, the novel 6 DOF KUKA robot featured a parallel linkage for its second and third axes - became a workhorse for the automotive industry.
- (b) The six axis PUMA was inspired by the dexterity of a human arm in 1979 by Unimation, due to its versatility and ease of use, as a reference in robotics research for many years.

Industrial Robot in Manufacturing

❑ Short History of Industrial Robots in Manufacturing



Parallel robots are slowly diffusing into various fields of industrial application (a) the Neos Tricept 600, (b) Fanuc F-200iB. (c) Adept Quattro

(a) In 1992, Neos Robotics combine the stiffness of machine tools with the dexterity of a robot for heavy-duty applications such as in friction stir welding or machining of aluminum for the aircraft industry.

(b) The Fanuc F-200iB introduced in 2002 is a 6-DOF parallel robot particularly designed for welding gun handling, deflashing, or for assembly tasks (100 kg payload, 0.1mm accuracy) in automotive assembly processes.

(c) the Adept Quattro is suited for high-speed applications in packaging, manufacturing, assembly, and material handling.

Industrial Robot in Manufacturing

❑ Short History of Industrial Robots in Manufacturing



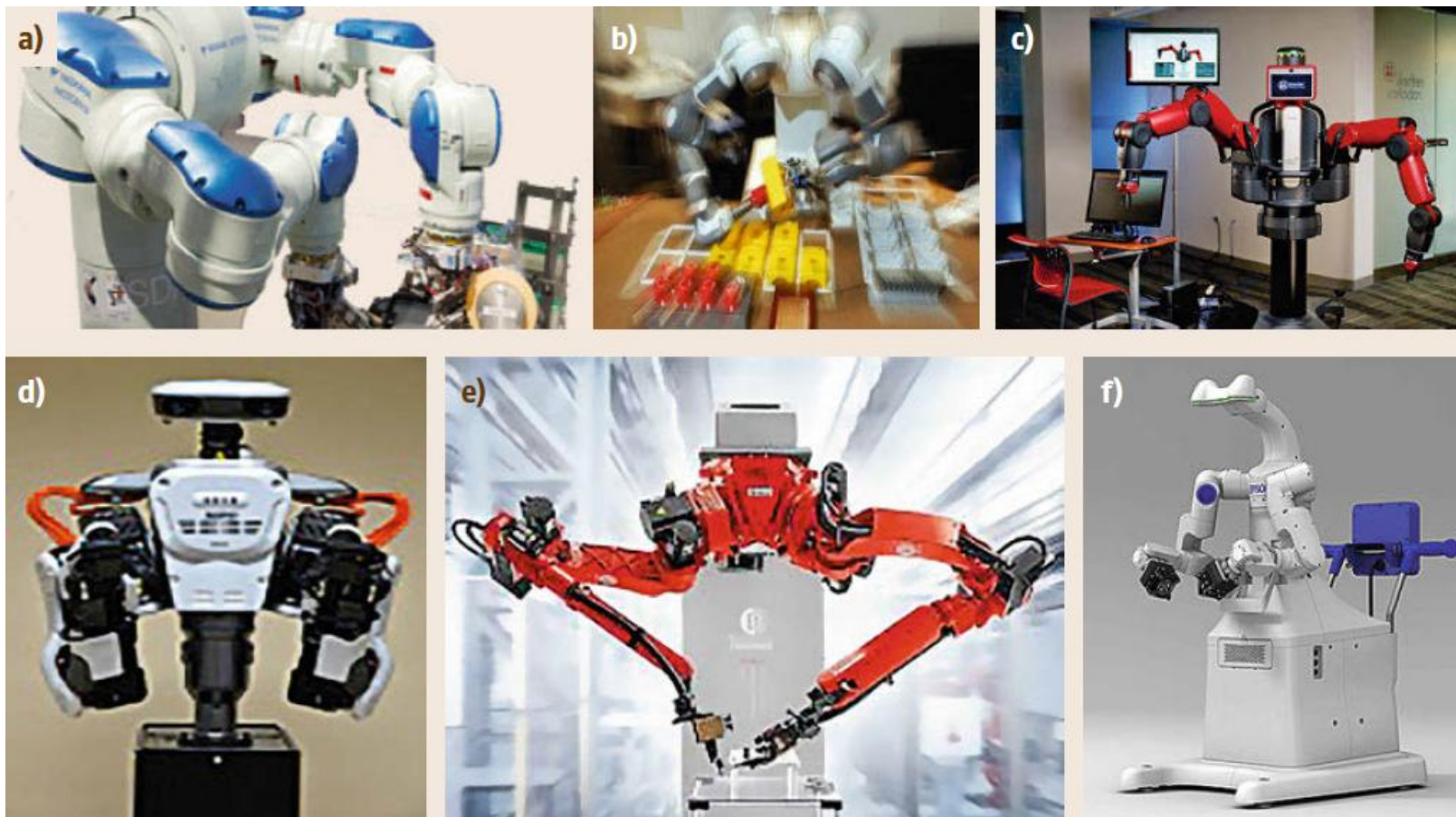
Mobile robots were introduced in the early 1980s for increased flexibility and reliability in factory logistics

(a) The MORO (1984) developed at Fraunhofer IPA was one of the first prototypes to combine a robot arm on a wirebound mobile platform which follows a wire buried in the floor.

(b) The KUKA omniRob features an omnidirectional platform and the LBR iiwa lightweight arm which form a highly kinematically redundant robot system.

Industrial Robot in Manufacturing

❑ Short History of Industrial Robots in Manufacturing



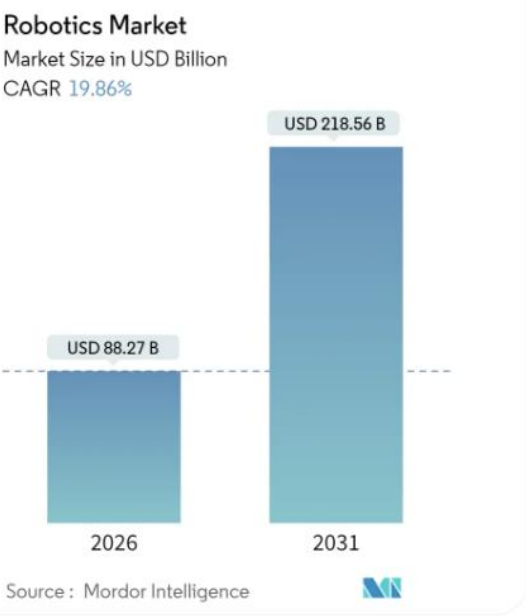
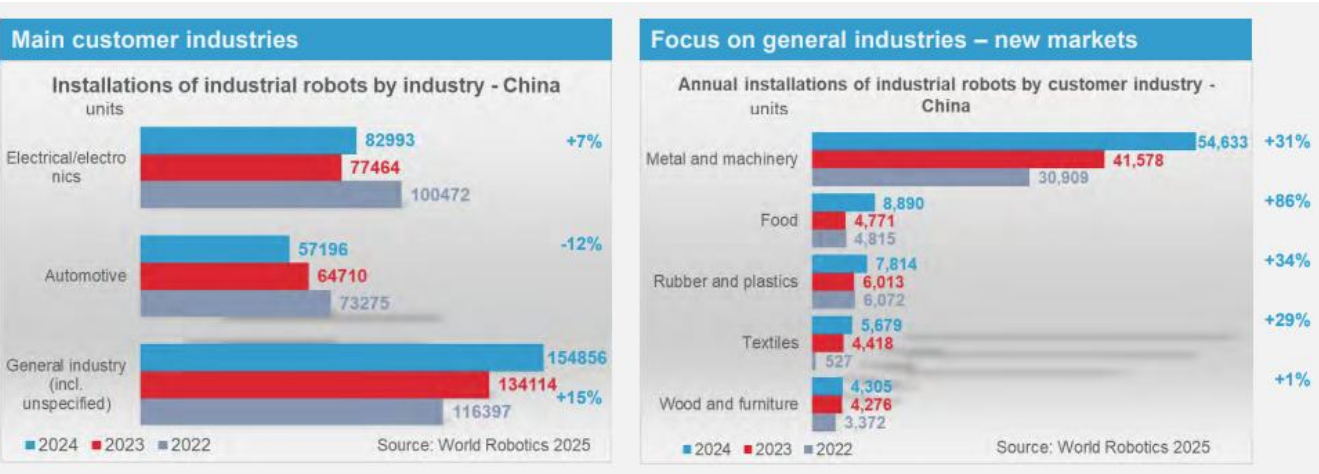
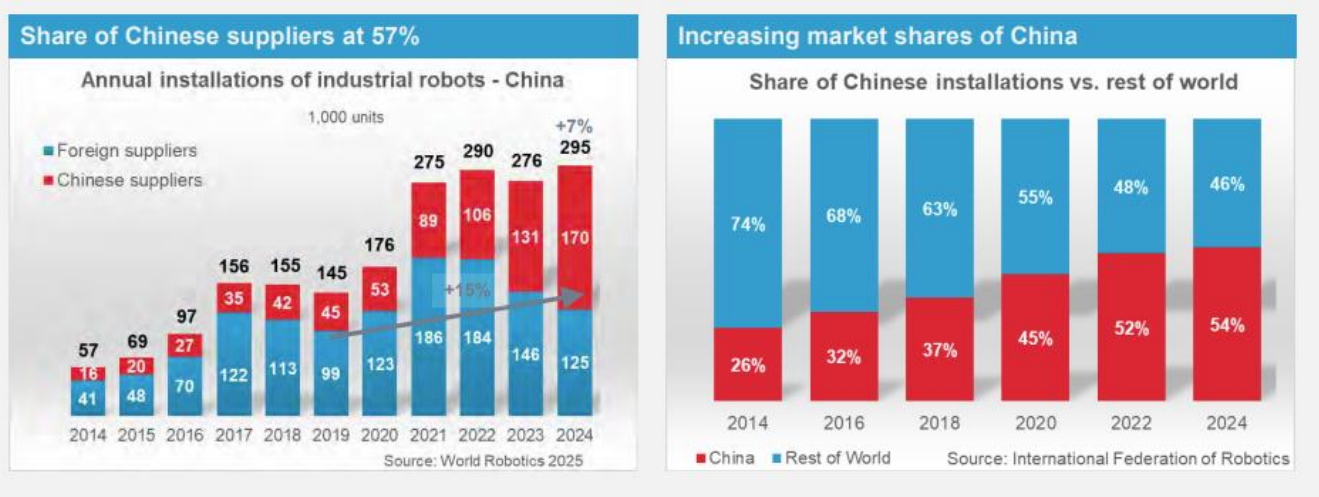
Examples of different designs of dual-arm robots

(a) Motoman, (b) ABB, (c) Rethink Robotics, (d) Kawada Industries, (e) COMAU, (f) Seiko Epson.



Industrial Robot in Manufacturing

Statistics of worldwide industrial robotics use: Growth in China comparable to global growth



Market Overview	
Study Period	2020 - 2031
Market Size (2026)	USD 88.27 Billion
Market Size (2031)	USD 218.56 Billion
Growth Rate (2026 - 2031)	19.86% CAGR
Fastest Growing Market	Middle East
Largest Market	Asia Pacific
Market Concentration	Medium
Major Players	
FANUC DENSO ABB YASKAWA KUKA	

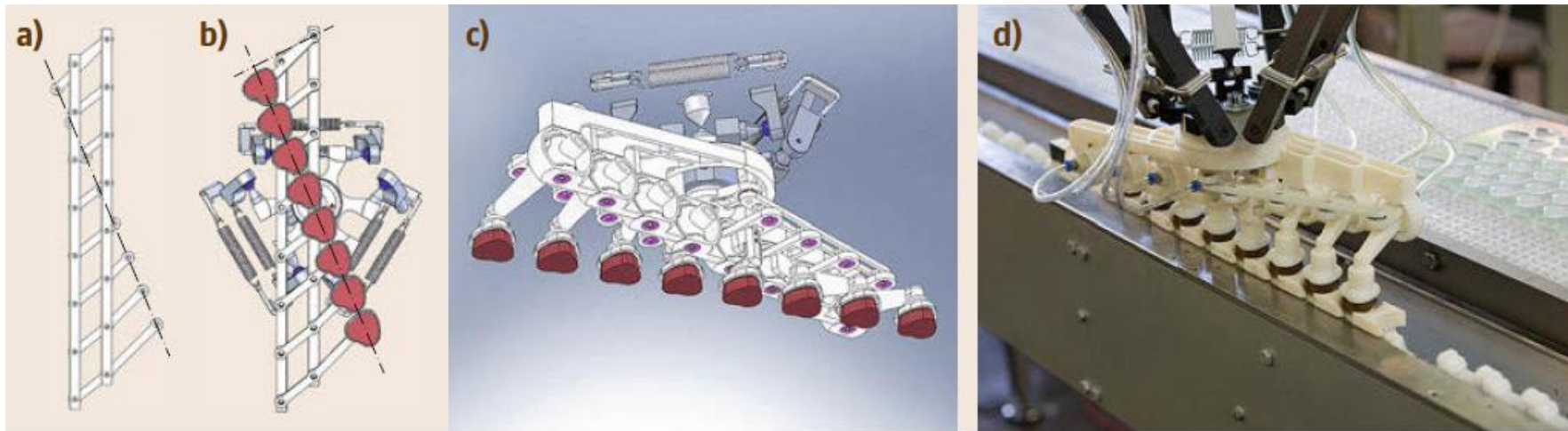
Global industrial robotics market

Industrial Robot in Manufacturing

❑ 1. Typical application in manufacturing - **Handling**

Handling in robotics comprises numerous processes such as grasping, transporting, packaging, palletizing, and picking.

Food Handling



Lightweight customized vacuum gripper through additive manufacturing.

Cookies are delivered continuously on a belt, grasped from the belt, and batches of eight are put on a blister matrix before final packaging.

The gripper's spacing is pneumatically actuated and its rotation through the parallel robot's central rotational axis.

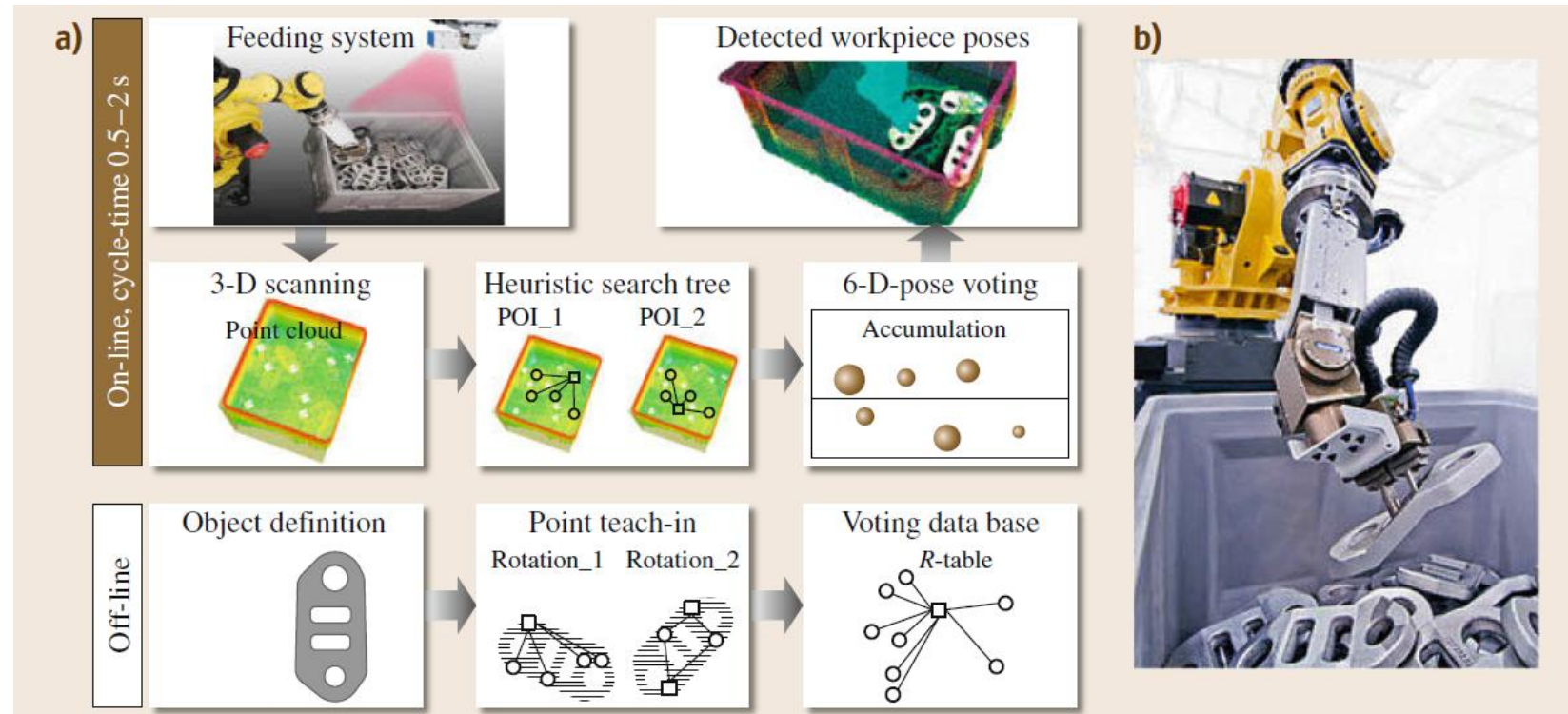
Industrial Robot in Manufacturing

❑ 1. Typical application in manufacturing - **Handling**

Handling in robotics comprises numerous processes such as grasping, transporting, packaging, palletizing, and picking.

Bin-Picking: The challenge of grasping partly or randomly ordered parts by robot has been referred to as bin-picking.

Bin-picking algorithms follow a typical sequence of steps: initial point cloud data acquisition, object detection, pose estimation, collision-free path and grasp planning, object grasping, and object placement.



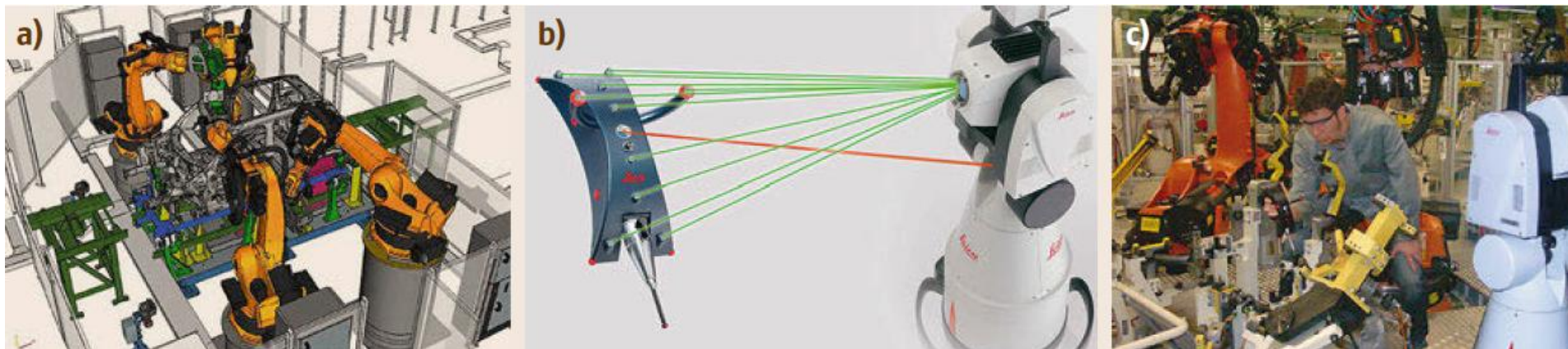
(a) Procedure of a bin-picking method; **(b)** gripper with additional degree of freedom for reaching deep into bins

Industrial Robot in Manufacturing

❑ 2. Typical application in manufacturing - **Welding**

Typical robot-based welding processes are spot welding, particularly in car body assembly, and gas-shielded metal arc welding (GMAW). With increasing compactness of laser sources and robot motion accuracy, laser welding is in the rise.

Spot Welding in the Automotive Industry with Offline Programming



Offline programming of a spot welding work cell

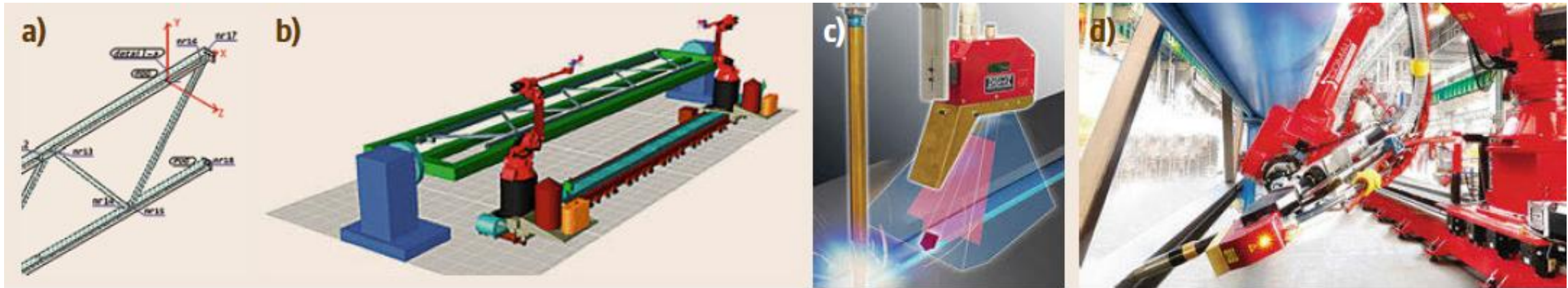
- (a) The robot work cell and the task execution are modeled.
- (b) Measure the spot welding location: the **laser tracker** is utilized to accurately measure the location of spot welding.
- (c) A tracker in use for interactively measuring the geometry of the robot work cell.

Industrial Robot in Manufacturing

□ 2. Typical application in manufacturing - **Welding**

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Arc Welding in Metal Construction



GMAW welding of building trusses

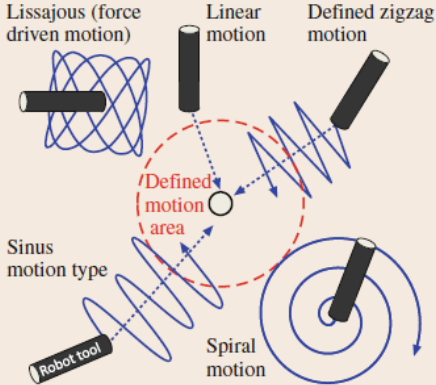
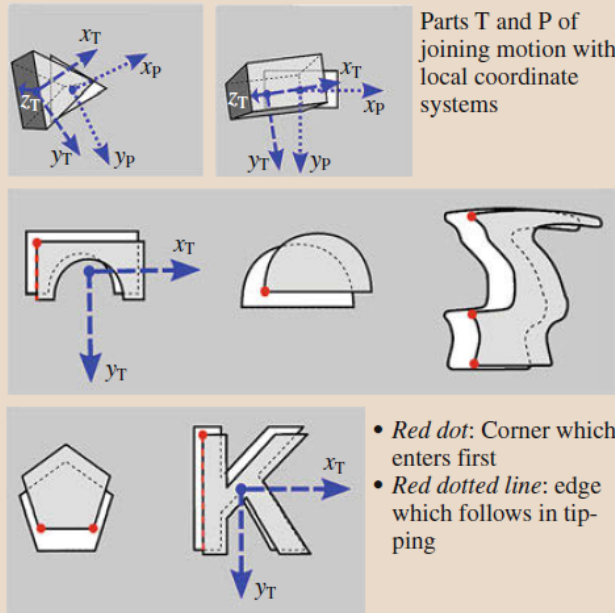
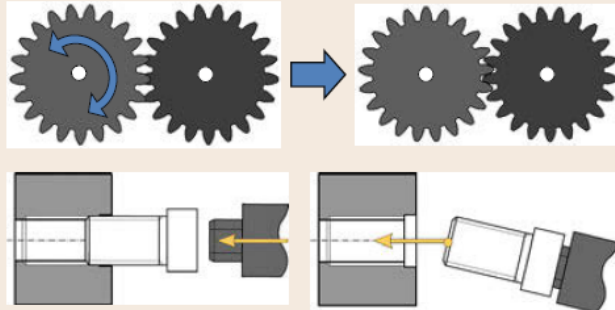
- (a) shows the CAD drawing of a steel truss with relevant information for welding process.
- (b) two welding robots working simultaneously on the truss.
- (c) the laser-based seam finding and tracking sensors.
- (d) the welding robot.

Industrial Robot in Manufacturing

3. Typical application in manufacturing - Assembly

Assembly in manufacturing describes the combination of subsystems or components to systems of a higher complexity through joining, comprises four process groups: joining, handling, controlling, and auxiliary processes (cleaning, adjustment, marking, etc.)

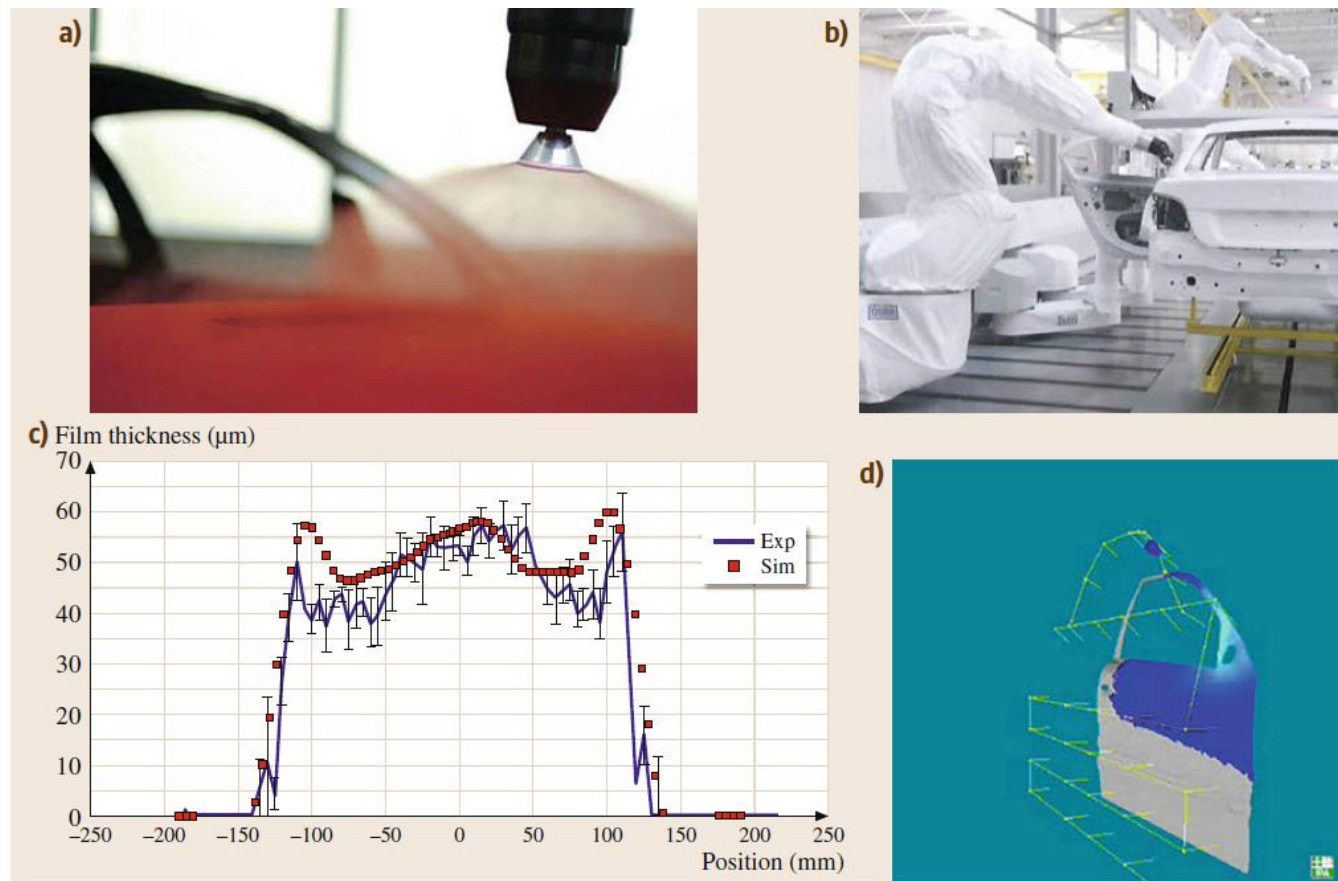
Table - Assembly subprocesses

Assembly subprocess	Characterization	Principle
<i>search</i> : search module which supports several search motions or search strategies	Search motion type examples: - Linear - Zigzag - Spiral - Sinus - Lissajous Commanding the search motion generation: - Position-based trajectory - Force-based trajectory - Combination of position and force-based trajectory	
<i>peg in hole</i> : execution of typical part joining motions	Reduction of arbitrary joining part types to three abstract planar types: - Round (arbitrary axial orientation) - Triangular (defined axial orientation) - Rectangular (additionally defined workpiece coordination system) Strategy of triangular types is common for numerous workpiece contours: - Orientation is given, tipping is executed in one step; object pivots around one corner	 Parts T and P of joining motion with local coordinate systems - Red dot: Corner which enters first - Red dotted line: edge which follows in tipping
<i>gear, search</i> : toothed gear joining motion and <i>screw-in</i> motion	- Meshing in toothed gears: Torque oscillations about gear axis and linear forward motion - Screw-in motion: fixed torque for torque-based screwing - Angle-controlled tightening required in most screwing process applications: command fixed torque, then turn by a defined angular increment.	

Industrial Robot in Manufacturing

□ 4. Typical application in manufacturing - **Painting**

Hazardous working conditions for human operators, spray guns for robots have evolved dramatically for delivering uniform quality using as little paint and solvent as possible and also for switching between different paint colors.



(a) A paint gun atomizes the paint material at the edge of the rotating bell disk by centrifugal forces. All current paint materials can be used in car production.

(b) Multiple robots work in parallel for optimal throughput and accessibility of the car body, with respect to optimal coverage of the painted surface.

(c) Simulation of the painting process is critical for achieving highest yields (e.g., minimal overspray, uniform paint deposition, etc.)

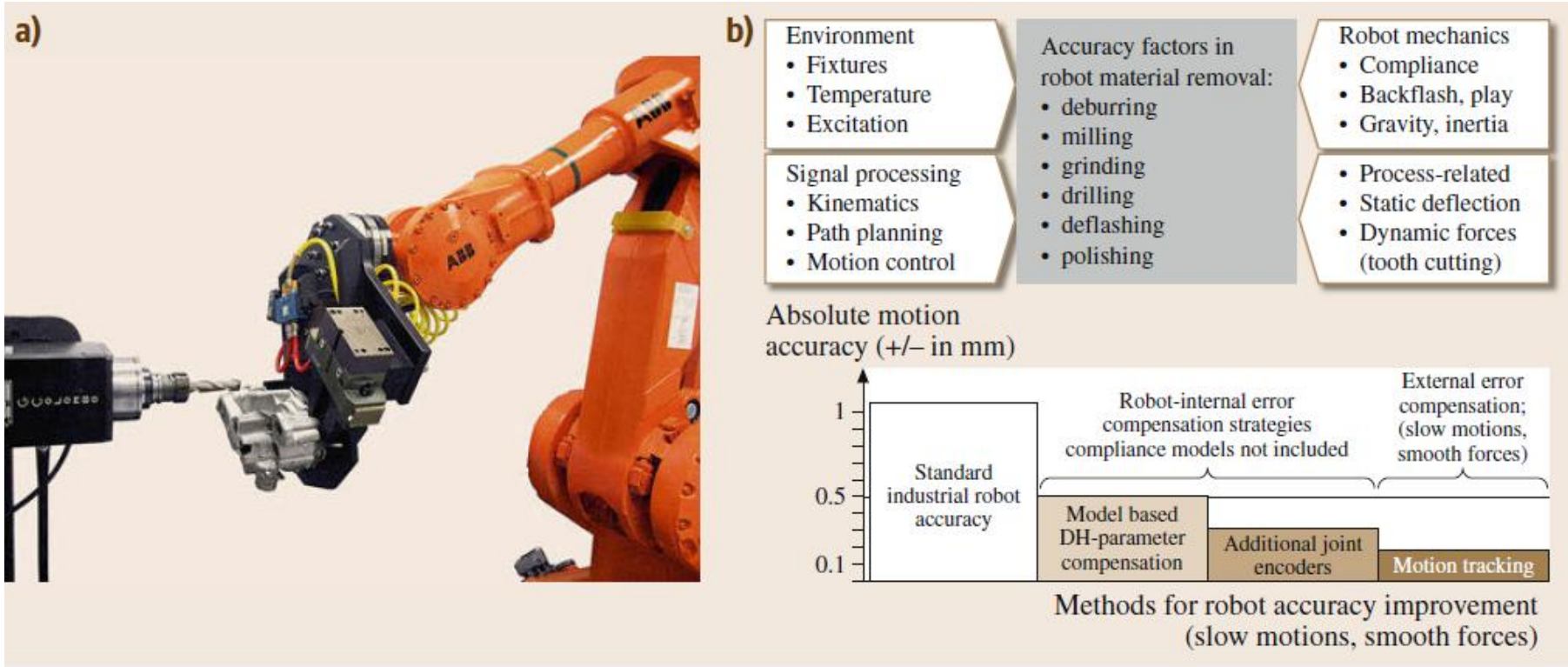
(d) Simulation result.

High-speed rotating atomizer and a multirobot work cell for car body painting

Industrial Robot in Manufacturing

5. Typical application in manufacturing - **Machining**

Material removal processes such as grinding, deburring, milling, and drilling are increasingly carried out by industrial robots with serial kinematics as they combine dexterity, versatility, and cost-effectiveness.



For small-to-medium-sized parts, a preferred configuration is to have the part grasped and guided by the robot relative to the fixed spindle. (a) The robot's gripper is mounted on a force–torque sensor to measure and limit process forces; A machine-learning technique is utilized to optimize the motions. (b) Influences on accuracy in robotic machining and methods for robot accuracy improvement, typical machining accuracies is 0.05mm or better.



Industrial Robot in Manufacturing

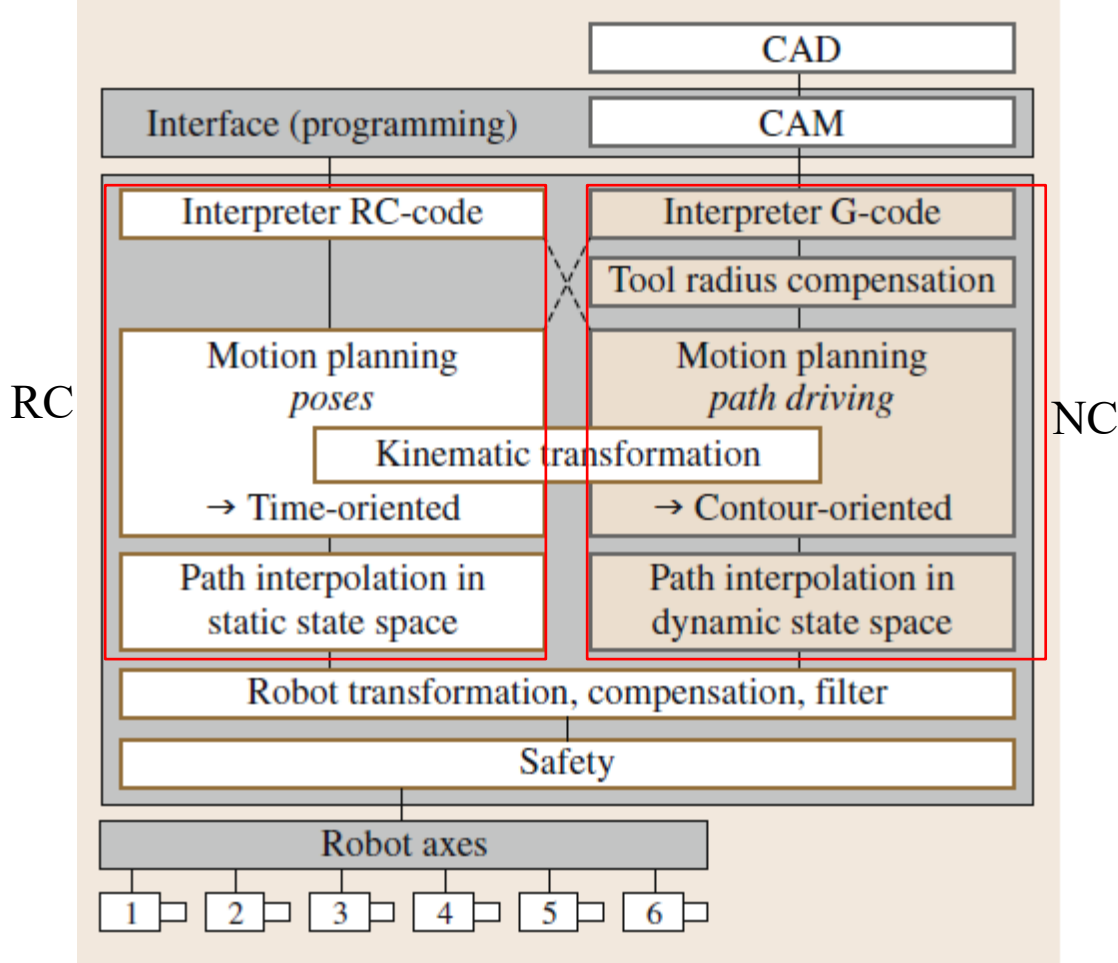
❑ 5. Typical application in manufacturing - **Machining**

Characteristics of computerized numerical control (CNC) and robot controller (RC)

Category	CNC	RC	Interpretation
Targeted application	Machining, material removal	Handling, assembly	A CNC machine tool is single purpose; generally robots are universal machines
Motions	Path-based, complex contour-oriented	Point- or path-based and motion time-oriented	Extended look-ahead of CNC controllers allows detailed path description and adaption on $\gg 100$ via points (< 10 via points in most robot controllers)
Programming	On-site programming based on standardized programming language (G-code), ISO (International Organization for Standardization) 6983, use of computer aided manufacturing (CAM) tools	On-site teaching (teach pendant, editor) based on supplier specific languages, use of typical robot simulation environments	Whereas robots are traditionally programmed manually on site, CNC controllers use CAM technology to generate complex paths automatically based on CAD data
Command reading	Online interpreter, continuous loading of instruction	Initial loading of programs, which are usually interpreted, sometimes compiled)	Program size in robot controllers limited by memory, CNC interprets programs online, may execute an unlimited number of commands

Industrial Robot in Manufacturing

5. Typical application in manufacturing - **Machining**



Increasingly CNC controllers are used for robots in material removal applications for taking advantage of the well-established off-line programming tools in the CNC world and for improving motion accuracy of the robot for complex 3-D contours.

Modern robot controllers integrate so-called numerical control (NC) kernels which share components of the robot controller such as user interfaces, kinematic transformations, and safety functions.

Structure of an NC-kernel integrated into a robot controller



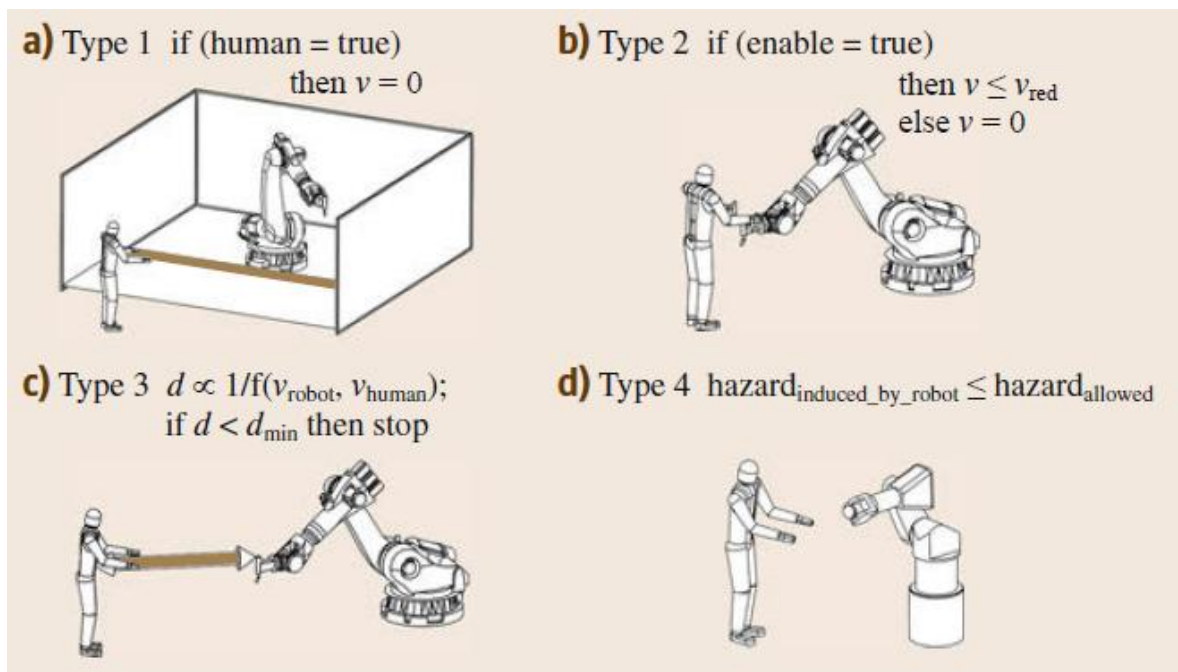
Industrial Robot in Manufacturing

❑ 6. Typical application in manufacturing – **Soft Human-Robot Collaboration**

Allows the combination of typical strengths of robots with some of the capabilities of humans.

- **Strengths of industrial robots** : high stamina, high payload capacity, precision, and repeatability.
- **Strengths of human workers**: flexibility for new production tasks, creative problem-solving skills, and the ability to react to unforeseen situations.

Industrial robots have a significant **potential to harm humans**. Standards for designing and operating industrial robot automation systems have been introduced.



Modes of human–robot collaboration according to ISO-10218.

(a) **Safety-rated monitored stop**: stop on access with automatic task resumption.

(b) **Hand-guiding**: direct physical interaction between human and robot with full control of the human over the robot movement.

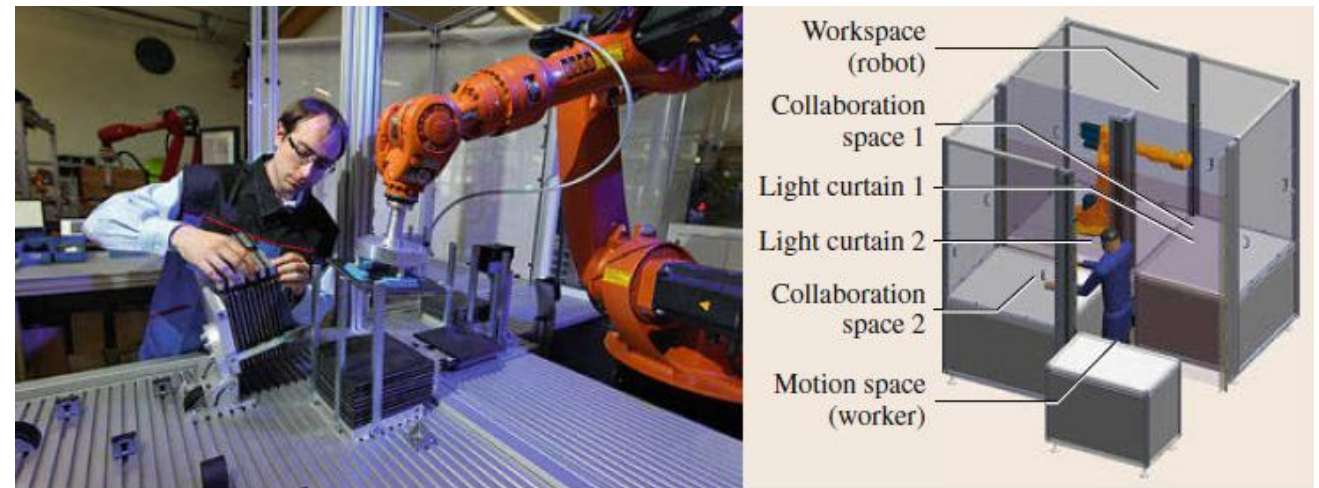
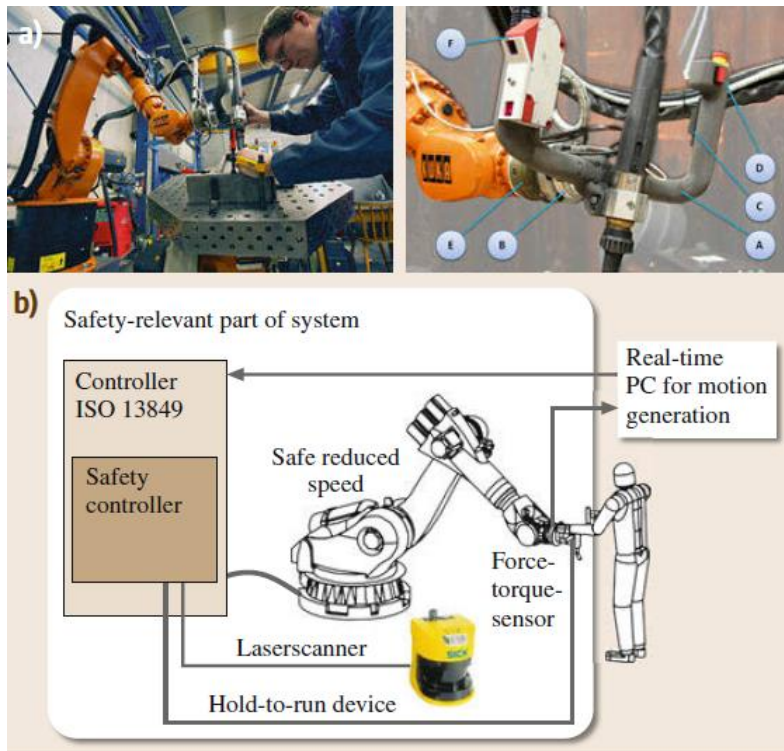
(c) **Separation and speed monitoring**: relative speed between robot and human as well as their distance are actively monitored.

(d) **Power and force limiting**: achieved by appropriately limiting collision forces through the design of the robot system.

Industrial Robot in Manufacturing

❑ 6. Typical application in manufacturing – **Soft Human-Robot Collaboration**

Examples of Human–Robot Collaboration



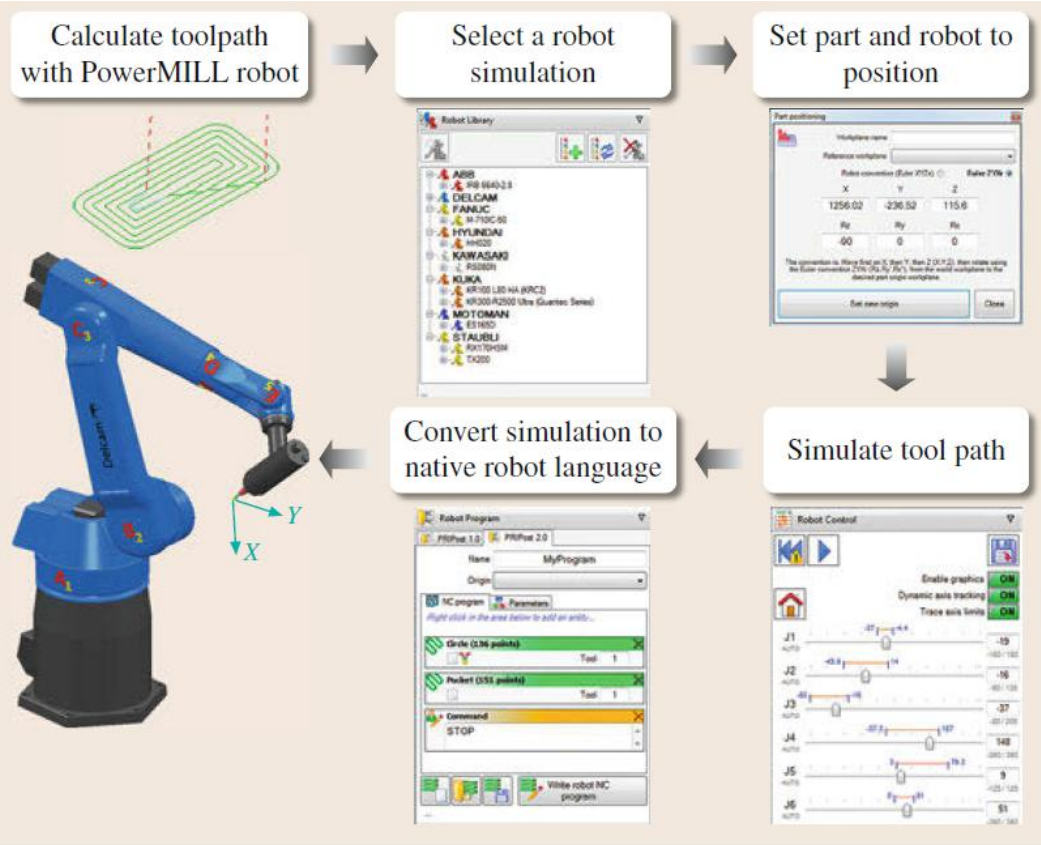
The robot's work cell is divided into three segments by two light curtains triggering in which area the robot works at lower velocity when collaborating with the worker in the collaboration spaces

Intuitive instruction of a welding robot by lead through programming: the worker guides the end effector of robot

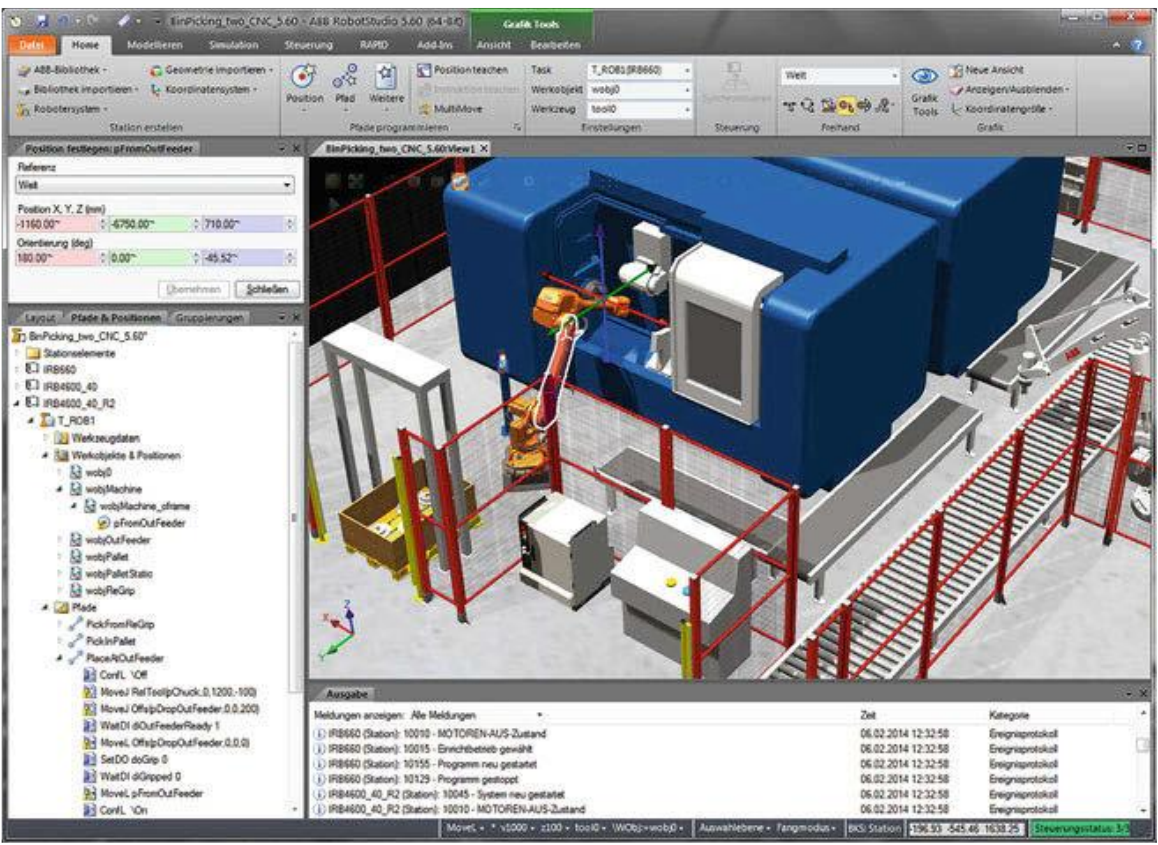
Industrial Robot in Manufacturing

7. Typical application in manufacturing – Robotic programming

Generate instructions that is quite explicit and motion oriented for robot.



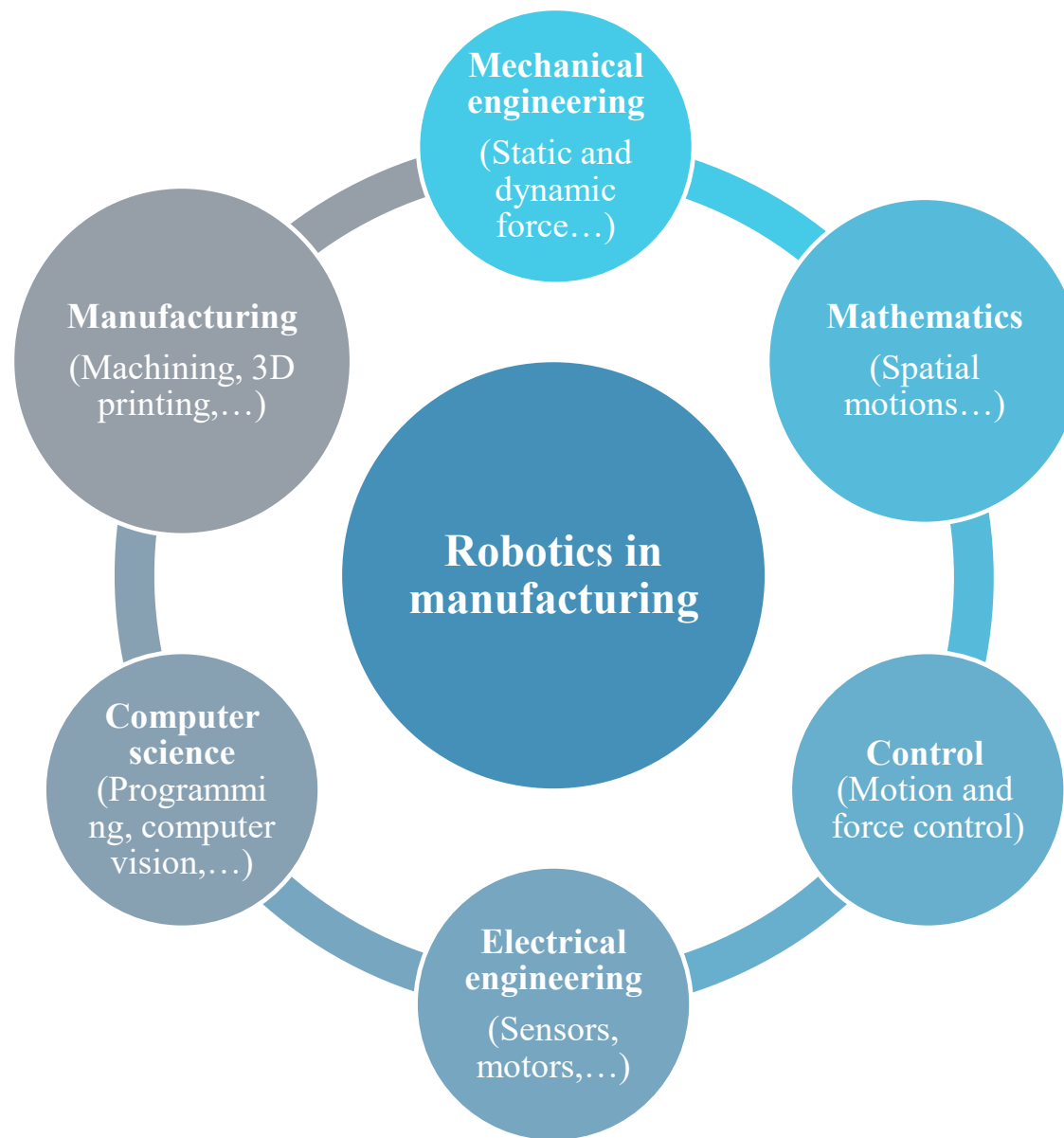
Workflow according to the PowerMILL software from Delcam with its robot module



Robot programming environment, with a virtual controller for each robot.

Inter-disciplinary Nature of Robotics

- **Mechanical engineering**
- **Mathematics**
- **Control**
- **Electrical engineering**
- **Computer science**
- **Manufacturing**
- ...





- **Physical, Analytic & Generative AI**

- Physical AI coined as the new term
- Generative AI is opening up new applications

- **Single Purpose Humanoids**

- Few commercial deployments announced
- True multipurpose humanoids are far off yet





Q&A